

Coupling Independence Implies Zero-Freeness

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Abstract

For $\Delta \geq 2$ and $q \geq 11\Delta/6$, we prove that the antiferromagnetic q -state Potts partition function on finite simple graphs of maximum degree at most Δ has no Fisher zeros in a graph-uniform complex neighbourhood of $[0, 1]$. For $q > 11\Delta/6$, the proof establishes coupling independence throughout $[0, 1]$ using a soft version of Vigoda’s flip dynamics. Our main tool is a separator-shell transfer theorem for the Potts model on induced-subgraph-closed classes of graphs of maximum degree at most Δ , with $q \geq \Delta + 1$. It yields a graph-uniform zero-free neighbourhood of $[0, 1]$ from Hamming coupling independence at 0 and a uniform coupling-independence bound on each interval $[\delta, 1]$, $\delta \in (0, 1]$. We also obtain zero-free Lee–Yang polydiscs around the uniform field for vertex- and edge-colour fields. For Boolean Holant problems on bounded-degree graphs whose signatures come from a fixed finite family of log-concave symmetric signatures f with $f(0) > 0$, such as b -matchings, we obtain graph-uniform zero-free polytubes around every bounded box of nonnegative activities; their union is an open zero-free neighbourhood of the nonnegative orthant. An appendix summarizes further coupling-independence inputs and the zero-free regions they yield.

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1 Introduction

For a finite simple graph $G = (V, E)$ and a positive integer q , write $[q] = \{1, \dots, q\}$ and define the Potts partition function

$$Z_G(q; z) := \sum_{\sigma: V \rightarrow [q]} z^{m_G(\sigma)}, \quad m_G(\sigma) := |\{uv \in E : \sigma_u = \sigma_v\}|. \quad (1)$$

The antiferromagnetic interval $[0, 1]$ interpolates between proper graph colourings and independent spins: $Z_G(q; 0)$ counts proper q -colourings, while $Z_G(q; 1) = q^{|V|}$. The zeros in the complex edge activity z are called *Fisher zeros* [Fis65]. A zero-free neighbourhood of $[0, 1]$, uniform over graphs of maximum degree at most Δ , permits deterministic approximate counting by interpolation [Bar16, PR17].

We obtain such a neighbourhood for $q \geq 11\Delta/6$, the classical colouring threshold associated with Vigoda's flip dynamics [Vig00]. The proof extends colouring couplings to the soft Potts model and converts their real-variable control into complex zero-freeness through a general transfer theorem: coupling independence implies zero-freeness (Theorem 1.2).

1.1 Main result

We state the result for pinned partition functions, since conditioning is essential to the coupling argument. Let $\Lambda \subseteq V$, let $\tau: \Lambda \rightarrow [q]$ be a pinning (precolouring), and let $m_G(\tau)$ count monochromatic edges of $G[\Lambda]$. Define the *normalized pinned partition function*

$$\widehat{Z}_G^\tau(q; z) := \sum_{\substack{\sigma: V \rightarrow [q] \\ \sigma|_\Lambda = \tau}} z^{m_G(\sigma) - m_G(\tau)}. \quad (2)$$

This polynomial removes the contribution of pinned-only edges, those with both ends in Λ , from the usual pinned partition function $Z_G^\tau(q; z) := \sum_{\sigma: V \rightarrow [q], \sigma|_\Lambda = \tau} z^{m_G(\sigma)}$:

$$Z_G^\tau(q; z) = z^{m_G(\tau)} \widehat{Z}_G^\tau(q; z). \quad (3)$$

Unlike Z_G^τ , the polynomial $\widehat{Z}_G^\tau$ has no forced zero at $z = 0$ when τ is improper (that is, when $m_G(\tau) \geq 1$); for the empty pinning it equals $Z_G(q; z)$. We suppress q when fixed and, for $I \subseteq \mathbb{C}$, write $\mathcal{U}_\varepsilon(I) := \{z \in \mathbb{C} : \text{dist}(z, I) < \varepsilon\}$.

Theorem 1.1 (Potts zero-freeness near $[0, 1]$). *Let q, Δ be integers with $\Delta \geq 2$ and $q \geq 11\Delta/6$. There exists $\varepsilon = \varepsilon(q, \Delta) > 0$ such that, for every finite simple graph G of maximum degree at most Δ and every pinning τ ,*

$$\widehat{Z}_G^\tau(z) \neq 0 \quad \text{for every } z \in \mathcal{U}_\varepsilon([0, 1]).$$

Consequently, $Z_G^\tau(z) = z^{m_G(\tau)} \widehat{Z}_G^\tau(z)$ has no zeros in the same neighbourhood except, when $m_G(\tau) \geq 1$, a zero of multiplicity $m_G(\tau)$ at $z = 0$. In particular, Z_G , and Z_G^τ for every τ proper on $G[\Delta]$, have no zeros in $\mathcal{U}_\varepsilon([0, 1])$.

Table 1 places the theorem among earlier Fisher zero-freeness results for general graphs: it lowers the best previous threshold, $q \geq 1.998\Delta$ [BBR26], to $q \geq 11\Delta/6 \approx 1.833\Delta$. For integer q , the new condition is strictly weaker than the previous ones for $\Delta = 2$ and every $\Delta \geq 6$, and coincides with $q \geq 2\Delta$ for $3 \leq \Delta \leq 5$. These results concern zeros in z , with the integer q fixed.

Work	Parameters
Bencs, Davies, Patel, and Regts [BDPR21]	$q \geq e\Delta + 1$
Liu, Sinclair, and Srivastava [LSS25]	$\Delta \geq 3, q \geq 2\Delta$
Bencs, Berrekkal, and Regts [BBR26]	$q, \Delta \geq 3, q \geq 1.998\Delta$
This paper	$\Delta \geq 2, q \geq 11\Delta/6$

All rows give a graph-uniform zero-free neighbourhood of $[0, 1]$ for finite simple graphs of maximum degree at most Δ . The first work also allows independent complex edge activities.

Table 1: Fisher zero-freeness for the Potts model on general graphs.

For triangle-free graphs, Liu, Sinclair, and Srivastava show that, for every $\alpha > \alpha_* \approx 1.7632$, where $\alpha_* e^{-1/\alpha_*} = 1$, there is a constant $\beta = \beta(\alpha)$ such that, for every $\Delta \geq 3$, triangle-free graphs of maximum degree at most Δ have a zero-free neighbourhood of $[0, 1]$ whenever $q \geq \alpha\Delta + \beta$ [LSS25, Theorem 1.3]. Their proof uses the strong spatial mixing argument of Gamarnik, Katz, and Misra [GKM15].

For fixed (q, Δ) and rational $x \in [0, 1]$, Theorem 1.1 gives a fully polynomial-time approximation scheme (FPTAS) for $Z_G(q; x)$: apply Barvinok’s interpolation method [Bar16] from $Z_G(q; 1) = q^{|V|}$, computing the required low-order coefficients with the bounded-degree algorithm of Patel and Regts [PR17]. The radius ε can be chosen effectively: the coupling constants used are explicit (Theorem 4.1 and Corollary 4.13, and, at $x = 0$ in the equality case $q = 11\Delta/6$, a bound polynomial in $q\Delta$ from [CFG⁺25, Theorem 20]), and the bounded local systems used in the proof form a finite, effectively enumerable family.

For each fixed $x \in (0, 1]$ and $\Delta \geq 3$, an FPTAS also follows from the coupling bounds of Section 4 and the counting theorem of [CFG⁺25, Theorem 4], whose marginal-lower-bound hypothesis holds automatically at a fixed positive activity [CFG⁺25, Observation 10]; stronger counting thresholds at $x = 0$ are already known [CFG⁺25]. Our main conclusion is therefore the uniform exclusion of complex zeros throughout the antiferromagnetic interval $[0, 1]$.

1.2 Coupling independence implies zero-freeness

Vigoda’s bounded two-colour flips give rapid mixing for proper colourings when $q > 11\Delta/6$ [Vig00]. Chen, Feng, Guo, Zhang, and Zou use *coupling independence* to turn contractive colouring couplings into deterministic approximate counting [CFG⁺25]. Our transfer theorem takes Hamming coupling independence as its probabilistic input: for every pinning and every free site, the two conditional

Gibbs laws at edge activity x obtained by fixing that site to two different colours admit a coupling whose expected Hamming disagreement on the remaining free sites is bounded independently of the graph, uniformly over x in a given set (Definition 2.2).

Theorem 1.2 (Coupling independence implies zero-freeness). *Fix integers q, Δ with $\Delta \geq 2$ and $q \geq \Delta + 1$, and let $\mathcal{G}$ be a class of finite simple graphs of maximum degree at most Δ that is closed under taking induced subgraphs. Assume:*

- (i) $\mathcal{G}$ satisfies coupling independence at $x = 0$ with constant C_0 ;
- (ii) for every $\delta \in (0, 1]$, it satisfies coupling independence on $[\delta, 1]$ with a finite constant C_δ .

Then there is $\varepsilon > 0$ such that, for every $G \in \mathcal{G}$ and every pinning τ ,

$$\hat{Z}_G^\tau(z) \neq 0 \quad \text{for every } z \in \mathcal{U}_\varepsilon([0, 1]).$$

Consequently, $Z_G^\tau(z) = z^{m_G(\tau)} \hat{Z}_G^\tau(z)$ has no zeros in the same neighbourhood except, when $m_G(\tau) \geq 1$, a zero of multiplicity $m_G(\tau)$ at $z = 0$.

Unlike the general counting theorem of [CFG⁺25, Theorem 4], the transfer uses no lower bound on nonzero conditional marginals. At a fixed positive activity, such a marginal lower bound holds automatically for bounded-degree Potts models [CFG⁺25, Observation 10], but it need not remain bounded away from zero as $x \downarrow 0$: a colour that appears on $k \geq 1$ pinned neighbours of a free vertex v has conditional marginal of order x^k at v . Theorem 1.2 is proved in Section 3. To deduce Theorem 1.1, we verify the hypotheses of Theorem 1.2 for all graphs of maximum degree at most Δ by extending Vigoda’s flip coupling to positive Potts activities (Section 4). For $q > 11\Delta/6$, the resulting coupling bound is uniform on all of $[0, 1]$. At equality, it is uniform on each $[\delta, 1]$; we combine these bounds with the known hard-colouring bound at $x = 0$ [CFG⁺25, Theorem 20].

Remark 1.3 (Beyond Potts). Theorem 1.2 is written in Potts notation to keep the statement concrete. Its proof uses a finite local state space, bounded local interactions, and the Hamming coupling bound. Extending it to another model also requires closure under the relevant pinnings and uniform control of the local factors, including those that vanish at a real base point. Section 5 verifies these properties for colour fields and log-concave Holant problems.

1.3 Related work

The zero-freeness proofs of Liu, Sinclair, and Srivastava and of Bencs, Berrekkal, and Regts perturb real ratio-recursion estimates to nearby complex activities [LSS25, BBR26], while that of Bencs, Davies, Patel, and Regts traps the complex ratios directly [BDPR21]. Recent work also obtains zero-freeness by complexifying Markov-chain arguments [LWYY25, LY26]. Here the complex-analytic step instead uses real couplings through a separator comparison.

Classical work of Dobrushin and Shlosman showed that, for lattice systems on $\mathbb{Z}^d$, a strong form of spatial mixing is equivalent to complete analyticity, which includes nonvanishing of finite-volume partition functions under small complex perturbations [DS85, DS87]. For bounded-degree graph families, the ratio recursions underlying correlation-decay analyses have been studied at complex parameters, through contraction or complex-dynamics arguments, to prove zero-freeness [PR19, LSS19, PR20, SS21, LSS25, CSS26]. For the hard-core model on bounded-degree induced-subgraph-closed families, Peters, Regts, and Reppekus show that a strengthening of strong spatial mixing implies zero-freeness [PRR26].

Conversely, zero-freeness under pinnings implies strong spatial mixing [Reg23, SY24], or an asymptotic form of it at superlogarithmic distances [Gam23]; for the ferromagnetic Ising model,

zero-freeness without pinnings yields a Weitz-type FPTAS throughout the Lee–Yang zero-free region, without requiring strong spatial mixing [SS26b].

Multivariate zero-freeness provides criteria for spectral independence [CLV24]. Under the hypotheses of the relevant mixing theorems, spectral-independence bounds yield rapid mixing of the corresponding dynamics [ALOG20]. Coupling independence was formalized by Chen and Zhang for Boolean distributions as a criterion for spectral independence [CZ23], and Chen and Feng used it to bound mixing times of multi-spin systems [CF24]; Chen, Feng, Guo, Zhang, and Zou showed that, with a marginal lower bound, it yields deterministic approximate counting on bounded-degree graphs [CFG⁺25]. For the Potts model, Theorem 1.2 shows that coupling independence also implies zero-freeness.

1.4 Proof overview

We begin with a simple averaging observation, also used by Chen, Feng, Guo, Zhang, and Zou in the proof of [CFG⁺25, Theorem 12]. If a coupling has expected total Hamming disagreement at most C , then among L disjoint distance shells around a root there is one with expected disagreement at most C/L . Taking L sufficiently large gives a separator with a bounded interior and a small coupling cost on its boundary.

Fix a real base point $x \in (0, 1]$, a pinning τ , and a free root v , and write $\tau^{v=c}$ for τ extended by $v \mapsto c$. Summing over the colouring ξ of such a shell writes each conditional partition-function response as an average under the *real* shell marginal ν_c :

$$\frac{\widehat{Z}_G^{\tau^{v=c}}(z)}{\widehat{Z}_G^{\tau^{v=c}}(x)} = \mathbb{E}_{\xi \sim \nu_c} [e^{h(\xi) + \eta_c(\xi)}].$$

Here $h(\xi)$ is the logarithmic response of the exterior, which does not depend on c , and $\eta_c(\xi)$ is the logarithmic response of the bounded interior, which is small when z is close to x . The exterior response itself need not be small; the argument uses only its variation across shell colourings. Induction makes h Lipschitz in the shell colouring, with a small constant A for Hamming distance. For two root colours a, b , our complex-average stability lemma (Lemma 3.1) then gives, with $W_{1,\text{Ham}}$ the Wasserstein distance for the Hamming metric,

$$\left| \text{Log} \frac{\mathbb{E}_{\nu_a} [e^{h + \eta_a}]}{\mathbb{E}_{\nu_b} [e^{h + \eta_b}]} \right| \leq 2A W_{1,\text{Ham}}(\nu_a, \nu_b) + 4 \max\{\|\eta_a\|_\infty, \|\eta_b\|_\infty\},$$

provided the oscillation of h and both perturbations are at most $1/8$. The left-hand side is the quantity that the induction bounds by A . On the chosen shell, $W_{1,\text{Ham}}(\nu_a, \nu_b) \leq C/L$, and for z close to x , the perturbations are at most $A/8$, so the right-hand side is at most $2AC/L + A/2 < A$ once $L \geq 64C$. The bound therefore reproduces itself, and the colour recursion (6) gives nonvanishing. At $x = 0$ this representation breaks down: some interior weights vanish there, and the configurations they forbid acquire weight under a complex perturbation. We control them as an additive error, obtaining a disc at zero that joins the positive-activity neighbourhoods.

To construct the real couplings, we use, for the colours c, c' at the two ends of an edge, the identity

$$x \mathbf{1}_{[c=c']} + \mathbf{1}_{[c \neq c']} = x + (1 - x) \mathbf{1}_{[c \neq c']}$$

to express each soft Potts interaction as a mixture of an unconstrained edge and a hard inequality. Conditioning on the random constraints gives a hard list-colouring instance. Two colourings that differ at a single vertex may induce different hard instances, so the usual flip coupling on a fixed

instance does not apply directly. We couple the constraint sets and construct a conditional coupling of the bounded flips that allows differences in the edges incident to the disagreement vertex and in its list; we then average the contraction estimate and apply stationary comparison [CFG⁺25, Lemma 19]. The new ingredients are this soft flip coupling and the complex stability argument converting coupling independence into nonvanishing.

Organization. Section 2 fixes notation and defines coupling independence. Section 3 proves Theorem 1.2, and Section 4 proves the soft flip coupling bound (Theorem 4.1) and Theorem 1.1. The same shell method also yields Lee–Yang zero-freeness for proper vertex and edge colourings under independent colour fields (Section 5.1), and, for Boolean Holant problems on bounded-degree graphs with signatures from a fixed finite family of log-concave symmetric signatures f satisfying $f(0) > 0$, graph-uniform zero-free polytubes around every bounded box of nonnegative activities (Section 5.2). Appendix A summarizes further coupling-independence inputs, mostly from the companion article [SS26a], and the zero-free regions they yield, including the general-graph threshold $q \geq 1.809\Delta$ for $\Delta \geq 125$.

2 Preliminaries

2.1 Pinnings and partition functions

Throughout, q and Δ are positive integers, $[q] = \{1, \dots, q\}$, $x \in [0, 1]$ is a real base point, and $z \in \mathbb{C}$ is the complex edge-activity variable. We write $\mathbf{1}_A$ for the indicator of an event A , and use the convention $0^0 = 1$. For a finite graph G , write

$$\Delta(G) := \max_{v \in V(G)} \deg_G(v)$$

for its maximum degree; the symbol Δ denotes a fixed degree bound. Let

$$\mathcal{G}_\Delta := \{G : G \text{ is finite, simple, and } \Delta(G) \leq \Delta\},$$

and, for $g \geq 3$, let

$$\mathcal{G}_{\Delta,g} := \{G \in \mathcal{G}_\Delta : \text{girth}(G) \geq g\}.$$

Both classes are closed under taking induced subgraphs. We use the convention that an acyclic graph, including the empty graph, has girth $+\infty$.

Fix $G = (V, E) \in \mathcal{G}_\Delta$. For a pinning $\tau : \Lambda \rightarrow [q]$, write

$$V^\tau := V \setminus \Lambda, \quad G^\tau := G[V^\tau],$$

and call G^τ the *free graph*. For $u \in V^\tau$ and $c \in [q]$, define the boundary count

$$b_u^\tau(c) := |\{p \in \Lambda : up \in E, \tau_p = c\}|. \quad (4)$$

Boundary counts record pinned neighbours separately, even when several have the same colour. The normalized pinned partition function from (2) has the factorized form

$$\hat{Z}_G^\tau(z) = \sum_{\varphi : V^\tau \rightarrow [q]} \prod_{\substack{uv \in E \\ \{u,v\} \not\subseteq \Lambda}} (z \mathbf{1}_{[(\tau \cup \varphi)_u = (\tau \cup \varphi)_v]} + \mathbf{1}_{[(\tau \cup \varphi)_u \neq (\tau \cup \varphi)_v]}). \quad (5)$$

Equivalently,

$$\widehat{Z}_G^\tau(z) = \sum_{\varphi: V^\tau \rightarrow [q]} \prod_{u \in V^\tau} z^{b_u^\tau(\varphi_u)} \prod_{uv \in E(G^\tau)} (z \mathbf{1}_{[\varphi_u = \varphi_v]} + \mathbf{1}_{[\varphi_u \neq \varphi_v]}).$$

Whenever $\widehat{Z}_G^\tau(x) > 0$, let $\mu_{G,x}^\tau$ be the Gibbs law on $[q]^{V^\tau}$:

$$\mu_{G,x}^\tau(\varphi) := \frac{1}{\widehat{Z}_G^\tau(x)} \prod_{u \in V^\tau} x^{b_u^\tau(\varphi_u)} \prod_{uv \in E(G^\tau)} (x \mathbf{1}_{[\varphi_u = \varphi_v]} + \mathbf{1}_{[\varphi_u \neq \varphi_v]}).$$

For $x > 0$, this is the usual conditional Gibbs law: removing the pinned-only edge contributions changes only the normalizing constant. At $x = 0$, it is the normalized hard pinned law, which remains meaningful even when τ is improper.

For $v \in V^\tau$ and $a \in [q]$, define the *child* pinning

$$\tau^{v=a} := \tau \cup \{v \mapsto a\}.$$

Then $V^{\tau^{v=a}} = V^\tau \setminus \{v\}$, and

$$\widehat{Z}_G^\tau(z) = \sum_{a \in [q]} z^{b_v^\tau(a)} \widehat{Z}_G^{\tau^{v=a}}(z). \quad (6)$$

The multiplier $z^{b_v^\tau(a)}$ accounts for the edges from v to Λ , which are pinned-only under $\tau^{v=a}$; the normalized function $\widehat{Z}_G^{\tau^{v=a}}$ omits them, so it has no forced zero at $z = 0$ even when this multiplier vanishes there.

2.2 Hard-colouring feasibility

A colour c is blocked at a free vertex u when $b_u^\tau(c) > 0$. The hard effective list of u is

$$L^\tau(u) := \{c \in [q] : b_u^\tau(c) = 0\}.$$

A colouring $\varphi: V^\tau \rightarrow [q]$ is *hard-feasible* if it is proper on G^τ and $\varphi_u \in L^\tau(u)$ for every $u \in V^\tau$.

Lemma 2.1 (Hard feasibility). *Assume $q \geq \Delta + 1$. For every $G \in \mathcal{G}_\Delta$, every pinning τ , and every $u \in V^\tau$,*

$$|L^\tau(u)| \geq \deg_{G^\tau}(u) + 1.$$

Consequently, $\widehat{Z}_G^\tau(0) > 0$.

Proof. At most $|N_G(u) \cap \Lambda|$ colours are blocked at u . Since

$$\deg_{G^\tau}(u) + |N_G(u) \cap \Lambda| = \deg_G(u) \leq \Delta,$$

we have

$$|L^\tau(u)| \geq q - |N_G(u) \cap \Lambda| \geq \deg_{G^\tau}(u) + 1.$$

A greedy list colouring of G^τ , in any vertex ordering, leaves an available colour at every step, so a hard-feasible colouring exists and $\widehat{Z}_G^\tau(0) > 0$. $\square$

2.3 Coupling independence and Wasserstein distance

We fix the Wasserstein notation and define coupling independence, the probabilistic hypothesis of Theorem 1.2.

Let $(\mathcal{X}, d)$ be a finite metric space. For probability measures ν_0, ν_1 on $\mathcal{X}$,

$$W_{1,d}(\nu_0, \nu_1) := \inf_{\pi \in \Pi(\nu_0, \nu_1)} \mathbb{E}_\pi[d(X, Y)], \quad (7)$$

where $\Pi(\nu_0, \nu_1)$ denotes the set of couplings of ν_0 and ν_1 , and X, Y are the two coordinate projections under π . On a finite space the infimum is attained. For configurations on a common site set,

$$\text{Ham}(\sigma, \sigma') := |\{u : \sigma_u \neq \sigma'_u\}|, \quad W_{1,\text{Ham}} := W_{1,d} \text{ with } d = \text{Ham}.$$

Definition 2.2 (Potts coupling independence). Let $C \in [0, \infty)$, $\mathcal{G} \subseteq \mathcal{G}_\Delta$, and $J \subseteq [0, 1]$. We say that $\mathcal{G}$ satisfies *C-coupling independence* on J if, for every $G \in \mathcal{G}$, every pinning τ , every $r \in V^\tau$, all $a, b \in [q]$, and every $x \in J$, the two pinned laws are well defined and

$$W_{1,\text{Ham}}(\mu_{G,x}^{\tau^{r=a}}, \mu_{G,x}^{\tau^{r=b}}) \leq C. \quad (8)$$

Both laws live on the common free set $V^\tau \setminus \{r\}$, where the Hamming distance is taken; at $x = 0$ they are the normalized hard pinned laws. When $J = \{x\}$, we say that $\mathcal{G}$ satisfies *C-coupling independence at x*.

For $q \geq \Delta + 1$, the normalized child law $\mu_{G,0}^{\tau^{r=a}}$ remains well defined even when a is blocked at r , by Lemma 2.1. In that case it is defined by the normalized child weights, rather than by conditioning the parent law $\mu_{G,0}^\tau$ on $\varphi_r = a$; in the recursion (6), only the multiplier $z^{b_r^\tau(a)}$ of the child term vanishes at $z = 0$.

3 From coupling independence to zero-freeness

This section proves Theorem 1.2. Section 3.1 proves a stability estimate for complex averages. Section 3.2 uses it in a separator induction on each interval $[\delta, 1]$: induction makes the exterior response Lipschitz on a separator shell, and coupling independence compares the two shell averages. At 0, some interior weights vanish, so Section 3.3 supplies an additive-error estimate that gives a disc around 0; Section 3.4 combines the two.

3.1 Stability of complex averages

The separator argument writes the ratio of a normalized partition function at z to its value at a real base point as $\mathbb{E}_\nu[e^{h+\eta}]$, where h is a common exterior response and η is a small local perturbation. We must compare two such averages when both the law and the perturbation change. The Lipschitz bound on h will turn a coupling of the two laws into a bound on the change of the exponent. The small oscillation of h , together with the small local perturbations, will prevent complex cancellation and allow us to compare logarithms.

Lemma 3.1 (Complex averages under nearby laws). *Let $(\mathcal{X}, d)$ be a finite metric space, let ν_0, ν_1 be probability measures on $\mathcal{X}$, and let $A \geq 0$, $\beta \in [0, 1/8]$. Suppose $h, \eta, \eta' : \mathcal{X} \rightarrow \mathbb{C}$ satisfy*

$$|h(\xi) - h(\xi')| \leq A d(\xi, \xi'), \quad (9)$$

$$\max_{\xi, \xi'} |h(\xi) - h(\xi')| \leq 1/8, \quad (10)$$

$$\|\eta\|_\infty, \|\eta'\|_\infty \leq \beta, \quad (11)$$

where $\|f\|_\infty = \max_\xi |f(\xi)|$. Then the averages $\mathbb{E}_{\nu_0}[e^{h+\eta}]$ and $\mathbb{E}_{\nu_1}[e^{h+\eta'}]$ are nonzero and, with Log of their ratio taken as the continuous branch that vanishes at $h = \eta = \eta' = 0$ on the convex set of triples (h, η, η') satisfying (9)–(11),

$$\left| \text{Log} \frac{\mathbb{E}_{\nu_0}[e^{h+\eta}]}{\mathbb{E}_{\nu_1}[e^{h+\eta'}]} \right| \leq 2A W_{1,d}(\nu_0, \nu_1) + 4\beta. \quad (12)$$

Proof. First subtract $h(\xi_*)$ for a fixed $\xi_* \in \mathcal{X}$. This multiplies both averages by the same nonzero factor, so it changes neither their ratio nor whether they vanish. Keeping the notation h for the centred function, (10) now gives $\|h\|_\infty \leq 1/8$.

Choose an optimal coupling (X, Y) , with $X \sim \nu_0$ and $Y \sim \nu_1$, so that $\mathbb{E}[d(X, Y)] = W_{1,d}(\nu_0, \nu_1)$. For $0 \leq t \leq 1$, interpolate the exponents on this common probability space:

$$u_t = t(h(X) + \eta(X)) + (1-t)(h(Y) + \eta'(Y)), \quad M(t) = \mathbb{E}[e^{u_t}].$$

The marginals identify the endpoints as

$$M(0) = \mathbb{E}_{\nu_1}[e^{h+\eta'}], \quad M(1) = \mathbb{E}_{\nu_0}[e^{h+\eta}].$$

Thus the required ratio is $M(1)/M(0)$.

We first check that this path stays away from zero. Each endpoint exponent has absolute value at most $1/8 + \beta \leq 1/4$, so the same is true of u_t . The power-series inequality $|e^w - 1| \leq e^{|w|} - 1$ yields

$$|M(t) - 1| \leq \mathbb{E}[|e^{u_t} - 1|] \leq e^{1/4} - 1 < 1, \quad |M(t)| \geq 2 - e^{1/4}.$$

Thus the path lies in the disc $D := \{w \in \mathbb{C} : |w - 1| < 1\}$, and both endpoint averages are nonzero.

Next we bound how fast the average moves. The state space is finite, so differentiation under the expectation gives

$$M'(t) = \mathbb{E}[e^{u_t}(h(X) - h(Y) + \eta(X) - \eta'(Y))].$$

The Lipschitz condition bounds the first difference by $A d(X, Y)$, and the two perturbations contribute at most 2β . Since $|e^{u_t}| \leq e^{1/4}$, we obtain

$$|M'(t)| \leq e^{1/4} \mathbb{E}[A d(X, Y) + 2\beta] = e^{1/4}(A W_{1,d}(\nu_0, \nu_1) + 2\beta).$$

This is where the coupling cost enters the estimate.

Finally, the principal logarithm Log is analytic on D , with $\text{Log} 1 = 0$. The chain rule and the fundamental theorem of calculus give

$$\text{Log} M(1) - \text{Log} M(0) = \int_0^1 \frac{M'(t)}{M(t)} dt.$$

This difference is the logarithm specified in the lemma: both averages lie in D for every admissible triple, so the difference is continuous on the convex set of admissible triples and vanishes at $h = \eta = \eta' = 0$. Combining the derivative bound with the lower bound on $|M(t)|$, we conclude that

$$\begin{aligned} \left| \text{Log} \frac{M(1)}{M(0)} \right| &\leq \int_0^1 \frac{|M'(t)|}{|M(t)|} dt \\ &\leq \frac{e^{1/4}}{2 - e^{1/4}} (A W_{1,d}(\nu_0, \nu_1) + 2\beta) \\ &\leq 2A W_{1,d}(\nu_0, \nu_1) + 4\beta, \end{aligned}$$

where the last step uses $e^{1/4}/(2 - e^{1/4}) < 2$. □

In the diagram below, each arrow label bounds the modulus of the logarithm of the ratio of its endpoint averages. The horizontal solid arrow uses $\eta = \eta' = 0$. Along a vertical arrow the law stays fixed, so we take $X = Y$, and one endpoint has zero perturbation. The perturbation difference in the derivative bound is then at most β , giving 2β . The dashed arrow is (12). Thus coupling independence enters through the horizontal Wasserstein term, while the local perturbations account for the two vertical terms.

$$\begin{array}{ccc}
\mathbb{E}_{\nu_0}[e^h] & \xleftarrow{2A W_{1,d}(\nu_0, \nu_1)} & \mathbb{E}_{\nu_1}[e^h] \\
\downarrow 2\beta & & \uparrow 2\beta \\
\mathbb{E}_{\nu_0}[e^{h+\eta}] & \xleftarrow{2A W_{1,d}(\nu_0, \nu_1) + 4\beta} & \mathbb{E}_{\nu_1}[e^{h+\eta'}]
\end{array}$$

3.2 Positive-activity estimates

The estimates in this subsection are uniform on every compact interval $[\delta, 1] \subset (0, 1]$. Fix $G \in \mathcal{G}_\Delta$, a pinning τ , and a free vertex v . All graph distances in this subsection are measured in the free graph G^τ . For $r \geq 0$, write

$$B_r(v) := \{u \in V^\tau : \text{dist}_{G^\tau}(u, v) \leq r\}, \quad S_r(v) := \{u \in V^\tau : \text{dist}_{G^\tau}(u, v) = r\}.$$

For colours a, b and a real base point x , let $\nu_a^{(r)}$ and $\nu_b^{(r)}$ be the $S_r(v)$ -marginals of $\mu_{G,x}^{\tau^{v=a}}$ and $\mu_{G,x}^{\tau^{v=b}}$. If the two pinned laws admit a coupling of expected Hamming cost at most C , then, for every integer $L \geq 1$,

$$\sum_{r=2}^{L+1} W_{1,\text{Ham}}(\nu_a^{(r)}, \nu_b^{(r)}) \leq C. \quad (13)$$

Indeed, fix a single coupling (X, Y) of the two pinned laws with expected Hamming cost at most C . Projecting this coupling onto each shell gives

$$W_{1,\text{Ham}}(\nu_a^{(r)}, \nu_b^{(r)}) \leq \mathbb{E}[\text{Ham}(X_{S_r(v)}, Y_{S_r(v)})].$$

Since the shells are disjoint, the right-hand sides sum to at most $\mathbb{E}[\text{Ham}(X, Y)] \leq C$, which proves (13). Hence one of these L shells has cost at most C/L . If a shell is empty, the component of v has size bounded in terms of (Δ, L) and is treated as a finite local system.

Choose $r \geq 2$ and put

$$S := S_r(v), \quad U := B_{r-1}(v) \setminus \{v\}, \quad G' := G - B_{r-1}(v);$$

for $\xi: S \rightarrow [q]$, write $\tau_\xi := \tau \cup \xi$. For $c \in [q]$, split a colouring at S as follows. Among the edge factors present in the normalized child $\widehat{Z}_G^{\tau^{v=c}}$, assign to the exterior precisely those with an endpoint in $V^\tau \setminus B_r(v)$. Let $D_{c,\xi}(z)$ be the sum, over colourings of U , of the product of the remaining factors, evaluated with $v = c$ and $S = \xi$. In particular, factors from edges within S and from edges joining S to Λ belong to $D_{c,\xi}$. Edges joining v to Λ are pinned-only in the child, so they contribute no factor here. Since no edge crosses directly from U to the exterior, we obtain the *separator identity*

$$\widehat{Z}_G^{\tau^{v=c}}(z) = \sum_{\xi: S \rightarrow [q]} D_{c,\xi}(z) \widehat{Z}_{G'}^{\tau_\xi}(z). \quad (14)$$

At a real base point x , the shell marginal $\nu_c = \nu_c^{(r)}$ is

$$\nu_c(\xi) := \frac{D_{c,\xi}(x) \widehat{Z}_{G'}^{\tau_\xi}(x)}{\widehat{Z}_G^{\tau^{v=c}}(x)}. \quad (15)$$

The exterior instance (G', τ_ξ) does not depend on the root colour c , and changing one coordinate of ξ changes only one pinned colour in it; this is where the inductive one-vertex response bound enters.

Lemma 3.2 (Positive-activity Potts induction). *Fix integers q, Δ with $\Delta \geq 2$, let $\delta \in (0, 1]$ and $C \in [0, \infty)$, and let $\mathcal{G} \subseteq \mathcal{G}_\Delta$ be closed under taking induced subgraphs and satisfy C -coupling independence on $[\delta, 1]$. There are $\alpha, \varepsilon > 0$, depending only on (q, Δ, δ, C) , such that, for every $G \in \mathcal{G}$, every pinning τ , every $x \in [\delta, 1]$, and every $z \in \mathbb{C}$ with $|z - x| < \varepsilon$,*

$$\widehat{Z}_G^\tau(z) \neq 0,$$

and, for every $v \in V^\tau$ and $a, b \in [q]$,

$$\left| \text{Log} \frac{\widehat{Z}_G^{\tau^{v=a}}(z) / \widehat{Z}_G^{\tau^{v=a}}(x)}{\widehat{Z}_G^{\tau^{v=b}}(z) / \widehat{Z}_G^{\tau^{v=b}}(x)} \right| \leq \alpha. \quad (16)$$

The logarithm is continued from $z = x$.

Proof. **Local bounds.** Enlarging C if necessary, assume $C \geq 1$, and take $L = \lceil 64C \rceil$. Set

$$s := \Delta(\Delta - 1)^L, \quad N := 1 + \Delta \sum_{k=0}^{L-1} (\Delta - 1)^k.$$

For $2 \leq r \leq L + 1$, breadth-first bounds give $|S_r(v)| \leq \Delta(\Delta - 1)^{r-1} \leq s$ and $|B_{r-1}(v)| \leq N$. Choose

$$0 < \alpha \leq \min \left\{ \frac{\pi}{16}, \frac{1}{16s} \right\}.$$

Here L provides enough disjoint shells to reduce the coupling cost, s bounds a shell, N separates the bounded-component case, and α is the logarithmic and angular budget. Up to relabelling, only finitely many interior polynomials $D_{c,\xi}$ and polynomials of free components with at most N vertices occur: each is determined by a graph of maximum degree at most Δ on at most $N + s$ vertices, the root, the shell, the colours c, ξ , and boundary counts in $\{0, \dots, \Delta\}$. They are positive on the compact interval $[\delta, 1]$. We may therefore choose $\varepsilon > 0$, uniformly in $x \in [\delta, 1]$, so that they remain nonzero for $|z - x| < \varepsilon$, the normalized child quotients of components of size at most N satisfy (16), and

$$\left| \text{Log} \frac{D_{c,\xi}(z)}{D_{c,\xi}(x)} \right| \leq \frac{\alpha}{8}, \quad \left| \Im \text{Log} \frac{z^m}{x^m} \right| + \alpha < \frac{\pi}{2} \quad (0 \leq m \leq \Delta). \quad (17)$$

All logarithms continue from $z = x$.

Exterior response. Induct on the number of free vertices, proving zero-freeness and (16) simultaneously. Fix G, τ, v, a, b . If the free component of v has size at most N , cancel the other components; the uniform local choice of ε gives the response bound directly. If it is larger, (13) supplies a shell S with

$$W_{1,\text{Ham}}(\nu_a, \nu_b) \leq C/L.$$

Induction on the common exterior defines

$$h_\xi := \text{Log} \frac{\widehat{Z}_{G'}^{\tau_\xi}(z)}{\widehat{Z}_{G'}^{\tau_\xi}(x)}.$$

We write $h(\xi) := h_\xi$. When ξ and ξ' differ at one shell vertex w , put $\tau' := \tau \cup \xi|_{S \setminus \{w\}}$. Then G' is an induced subgraph of G , so $G' \in \mathcal{G}$; the instance (G', τ') has $|V^\tau| - |B_r(v)| + 1 < |V^\tau|$ free vertices, and its two one-vertex pinnings at w are exactly (G', τ_ξ) and $(G', \tau_{\xi'})$; pinned-only factors are omitted by (5). Thus the inductive response bound applies to w in (G', τ') . Both $h_\xi - h_{\xi'}$ and that response logarithm are analytic and vanish at $z = x$, so they are the same continued branch. Summing along a Hamming path therefore gives

$$|h_\xi - h_{\xi'}| \leq \alpha \text{Ham}(\xi, \xi') \leq s\alpha \leq \frac{1}{16}.$$

Comparison of shell averages. The bounded interior has normalized change

$$\eta_c(\xi) := \text{Log} \frac{D_{c,\xi}(z)}{D_{c,\xi}(x)}, \quad \|\eta_c\|_\infty \leq \frac{\alpha}{8}.$$

The separator identity is now exactly

$$\frac{\widehat{Z}_G^{\tau^{v=c}}(z)}{\widehat{Z}_G^{\tau^{v=c}}(x)} = \mathbb{E}_{\nu_c}[e^{h+\eta_c}]. \quad (18)$$

The hypotheses of Lemma 3.1, with $A = \alpha$ and $\beta = \alpha/8$, hold at every z' with $|z' - x| < \varepsilon$, and h, η_a, η_b vanish at $z' = x$. Composing the continuous branch of Lemma 3.1 with $z' \mapsto (h, \eta_a, \eta_b)$ therefore gives a continuous logarithm of the ratio in (16) that vanishes at $z' = x$, namely the logarithm continued from $z = x$. Thus Lemma 3.1 gives

$$\left| \text{Log} \frac{\widehat{Z}_G^{\tau^{v=a}}(z)/\widehat{Z}_G^{\tau^{v=a}}(x)}{\widehat{Z}_G^{\tau^{v=b}}(z)/\widehat{Z}_G^{\tau^{v=b}}(x)} \right| \leq 2\alpha C/L + \frac{\alpha}{2} < \alpha.$$

Parent nonvanishing. To deduce nonvanishing of the parent function $\widehat{Z}_G^\tau$, put

$$R_c := \frac{\widehat{Z}_G^{\tau^{v=c}}(z)}{\widehat{Z}_G^{\tau^{v=c}}(x)}$$

and choose any $a_0 \in [q]$. The colour recursion (6) gives

$$\widehat{Z}_G^\tau(z) = R_{a_0} \sum_a x^{b_v^\tau(a)} \widehat{Z}_G^{\tau^{v=a}}(x) \exp\left(\text{Log} \frac{z^{b_v^\tau(a)}}{x^{b_v^\tau(a)}} + \text{Log} \frac{R_a}{R_{a_0}}\right).$$

The coefficients are positive. By (17) and the response bound, every exponential has positive real part, so the sum lies in an open half-plane and is nonzero. This closes the induction. $\square$

3.3 Hard-endpoint estimates

At $z = 0$, a boundary weight may vanish. Hard-feasible colourings contribute positive weight at $z = 0$, while the term of every colouring that is not hard-feasible is divisible by z .

Lemma 3.3 (One-vertex comparison at $z = 0$). *Assume $q \geq \Delta + 1$ and put $H_* := q^\Delta$. For every $G \in \mathcal{G}_\Delta$, every pinning τ , every $y \in V^\tau$, and all colours a, b ,*

$$\widehat{Z}_G^{\tau^{y=a}}(0) \leq H_* \widehat{Z}_G^{\tau^{y=b}}(0). \quad (19)$$

Proof. Measure distance in G^τ , condition on $S_2(y)$, and cancel free components outside that of y . For each shell colouring compatible with the exterior hard constraints, the exterior factor is the same for the two colours at y . Only the at most Δ vertices in $S_1(y)$ remain to be coloured. Fix such a shell colouring ξ and a root colour $c \in \{a, b\}$. For every $u \in S_1(y)$, after forbidding c and the colours used by its neighbours in $S_2(y)$, the remaining list has size at least

$$\begin{aligned} |L^\tau(u)| - 1 - |N_{G^\tau}(u) \cap S_2(y)| &\geq \deg_{G^\tau}(u) + 1 - 1 - |N_{G^\tau}(u) \cap S_2(y)| \\ &= \deg_{G^\tau[S_1(y)]}(u) + 1. \end{aligned}$$

Here the equality uses that every $u \in S_1(y)$ is adjacent to y , and its other neighbours lie in $S_1(y) \cup S_2(y)$. Greedy list-colouring of $G^\tau[S_1(y)]$ therefore gives at least one extension for each of the two root colours. There are at most $q^{|S_1(y)|} \leq q^\Delta$ extensions. Comparing term by term over the shell proves the claim. $\square$

Proposition 3.4 (Potts disc around $z = 0$). *Fix integers q, Δ with $\Delta \geq 2$ and $q \geq \Delta + 1$, let $C_0 \in [0, \infty)$, and let $\mathcal{G} \subseteq \mathcal{G}_\Delta$ be closed under taking induced subgraphs and satisfy C_0 -coupling independence at $x = 0$. There is $\rho_0 = \rho_0(q, \Delta, C_0) > 0$ such that*

$$\widehat{Z}_G^\tau(z) \neq 0 \quad (|z| < \rho_0)$$

for every $G \in \mathcal{G}$ and every pinning τ .

Proof. Setup and shell choice. Enlarging C_0 if necessary, assume $C_0 \geq 1$, put $L = \lceil 64C_0 \rceil$, and set

$$s := \Delta(\Delta - 1)^L, \quad N := 1 + \Delta \sum_{k=0}^{L-1} (\Delta - 1)^k.$$

For every vertex v and $2 \leq r \leq L + 1$, these quantities bound $|S_r(v)|$ and $|B_{r-1}(v)|$, respectively. Choose

$$0 < \alpha \leq \min \left\{ \frac{\pi}{16}, \frac{1}{16s} \right\}.$$

These constants have the same roles as in the proof of Lemma 3.2: L yields a low-cost shell, s bounds its Hamming diameter, N isolates bounded components, and α is the logarithmic budget. The only new step is to separate the hard-feasible contribution from an additive error divisible by z . Below K denotes a positive constant depending only on (q, Δ, C_0) , and may increase from line to line. We prove, for a sufficiently small uniform ρ_0 and all $|z| < \rho_0$, zero-freeness and

$$\left| \text{Log} \frac{\widehat{Z}_G^{\tau^{v=a}}(z)/\widehat{Z}_G^{\tau^{v=a}}(0)}{\widehat{Z}_G^{\tau^{v=b}}(z)/\widehat{Z}_G^{\tau^{v=b}}(0)} \right| \leq \alpha \quad (20)$$

for every $G \in \mathcal{G}$, every pinning τ , every $v \in V^\tau$, and all a, b . The denominators are positive by Lemma 2.1. Throughout this proof, endpoint logarithms are the analytic branches continued from $z = 0$ and normalized there by $\text{Log} 1 = 0$. Induct on the number of free vertices. In the current inductive instance, fix G, τ, v , measure distance in G^τ , and use the sets $B_r(v), S_r(v)$ and the separator data $S, U, G', \tau_\xi, D_{c,\xi}$ from (14)–(15), now at the real base point $x = 0$.

If the free component containing v has at most N vertices, the other components cancel in (20). Each remaining component factor F satisfies $F(0) \geq 1$ by Lemma 2.1 and $|F(z) - F(0)| \leq q^N |z|$ for $|z| \leq 1$, so (20) holds, with these factors nonzero, whenever $|z| < \alpha/(4q^N)$; we take $\rho_0 \leq \alpha/(4q^N)$. For a larger component, choose a shell with $W_{1,\text{Ham}}(\nu_a, \nu_b) \leq C_0/L$.

Exterior response. Induction on the common exterior defines

$$h_\xi := \text{Log} \frac{\widehat{Z}_{G'}^{\tau_\xi}(z)}{\widehat{Z}_{G'}^{\tau_\xi}(0)}.$$

For shell colourings ξ, ξ' that differ only at a shell vertex w , regard all other shell vertices as part of the base pinning and temporarily unpin w . This realizes the two normalized exterior partition functions as the two one-vertex pinnings of the same induced instance; pinned-only factors are omitted by (5). Hence the inductive response bound (20) applies to this smaller instance. Branch uniqueness at $z = 0$ then identifies $h_\xi - h_{\xi'}$ with the inductive one-vertex response logarithm. Summing along a Hamming path gives

$$|h_\xi - h_{\xi'}| \leq \alpha \text{Ham}(\xi, \xi') \leq s\alpha \leq \frac{1}{16}.$$

Hard-feasible contribution and error control. Here ν_c is the shell marginal at $x = 0$. For each c , choose a hard-feasible colouring of the free vertices under $\tau^{v=c}$, and let ξ_c be its shell restriction. This witness supplies at least one hard interior extension, so

$$D_{c, \xi_c}(0) \geq 1, \quad \widehat{Z}_G^{\tau^{v=c}}(0) \geq \widehat{Z}_{G'}^{\tau_{\xi_c}}(0).$$

For a fixed interior colouring φ , its contribution is $z^{m(\varphi)}$ for some $m(\varphi) \geq 0$. After subtracting the value at 0, this contribution is zero if $m(\varphi) = 0$ and is $z^{m(\varphi)}$ otherwise. Since there are at most q^N interior colourings,

$$|D_{c, \xi}(z) - D_{c, \xi}(0)| \leq q^N |z| \quad (|z| \leq 1/2).$$

Splitting $D_{c, \xi}(z)$ into its value at 0 and its error in the separator identity gives

$$R_c := \frac{\widehat{Z}_G^{\tau^{v=c}}(z)}{\widehat{Z}_G^{\tau^{v=c}}(0)} = M_c + E_c, \quad M_c := \mathbb{E}_{\nu_c}[e^h],$$

where, explicitly,

$$E_c = \frac{\sum_\xi (D_{c, \xi}(z) - D_{c, \xi}(0)) \widehat{Z}_{G'}^{\tau_\xi}(0) e^{h_\xi}}{\widehat{Z}_G^{\tau^{v=c}}(0)}.$$

Changing the shell colours one at a time and applying Lemma 3.3 at most s times gives

$$\widehat{Z}_{G'}^{\tau_\xi}(0) \leq H_*^s \widehat{Z}_{G'}^{\tau_{\xi_c}}(0).$$

There are at most q^s shell colourings, and $|h_\xi - h_{\xi_c}| \leq 1/16$. The preceding bounds therefore give

$$|E_c| \leq q^{N+s} H_*^s e^{1/16} |z| |e^{h_{\xi_c}}|.$$

For the main term,

$$\begin{aligned} \left| \frac{M_c}{e^{h_{\xi_c}}} - 1 \right| &\leq \sum_\xi \nu_c(\xi) \left| e^{h_\xi - h_{\xi_c}} - 1 \right| \\ &\leq e^{1/16} - 1 < \frac{1}{4}. \end{aligned}$$

Hence

$$|M_c| \geq (2 - e^{1/16}) |e^{h_{\xi_c}}| > \frac{3}{4} |e^{h_{\xi_c}}|.$$

These two estimates give $|E_c/M_c| \leq K|z|$. Shrink ρ_0 , if necessary, so that $\rho_0 \leq 1/2$, $K\rho_0 \leq 1/2$, and $4K\rho_0 < \alpha/2$. Then

$$|\text{Log}(1 + E_c/M_c)| \leq 2K|z|,$$

with the logarithm continued from $z = 0$.

Apply Lemma 3.1 with $A = \alpha$ and $\eta = \eta' = 0$. Its hypotheses hold at every z' with $|z'| < \rho_0$, and h vanishes at $z' = 0$; as in the proof of Lemma 3.2, the branch in Lemma 3.1 is therefore the logarithm of M_a/M_b continued from $z = 0$. The lemma gives

$$\left| \text{Log} \frac{M_a}{M_b} \right| \leq 2\alpha C_0/L.$$

We have

$$\text{Log} \frac{R_a}{R_b} = \text{Log} \frac{M_a}{M_b} + \text{Log}(1 + E_a/M_a) - \text{Log}(1 + E_b/M_b).$$

Combining this identity with $L \geq 64C_0$ and $4K|z| < \alpha/2$, we find that the absolute value of its right-hand side is at most $\alpha/32 + \alpha/2 < \alpha$. This proves (20).

Parent nonvanishing. To close the induction, we use the parent recursion

$$\widehat{Z}_G^\tau(z) = \sum_c z^{b_v^\tau(c)} \widehat{Z}_G^{\tau^{v=c}}(z).$$

Choose a hard-feasible colouring σ and put $a_0 = \sigma_v$, so $b_v^\tau(a_0) = 0$. After division by R_{a_0} , the terms with $b_v^\tau(c) = 0$ have positive real part. The a_0 -term is exactly $\widehat{Z}_G^{\tau^{v=a_0}}(0)$, so their sum has real part at least this value. The remaining terms are divisible by z . Here $|R_c/R_{a_0}| \leq e^\alpha$, and Lemma 3.3 gives $\widehat{Z}_G^{\tau^{v=c}}(0) \leq H_* \widehat{Z}_G^{\tau^{v=a_0}}(0)$; hence the at most q remaining terms have total modulus at most $K|z| \widehat{Z}_G^{\tau^{v=a_0}}(0)$. A final shrink of ρ_0 makes this error smaller than the hard-feasible contribution and closes the induction. $\square$

3.4 Proof of the transfer theorem

Proof of Theorem 1.2. Let ρ_0 come from Proposition 3.4, set

$$\delta_0 := \min\{1/2, \rho_0/4\},$$

and apply Lemma 3.2 with $\delta = \delta_0$ and $C = C_{\delta_0}$. Let ε_+ be the resulting uniform radius and put $\varepsilon := \min\{\varepsilon_+, \delta_0\}$. Let $z \in \mathcal{U}_\varepsilon([0, 1])$, and let $x \in [0, 1]$ be a nearest point to z , so that $|z - x| < \varepsilon$. If $x \leq \delta_0$, then $|z| < 2\delta_0 < \rho_0$; otherwise $x \in [\delta_0, 1]$ and the positive-activity induction applies. The statement for Z_G^τ then follows from (3). $\square$

4 Coupling independence for $q \geq 11\Delta/6$

This section supplies the coupling input to Theorem 1.2 and proves Theorem 1.1. Above the threshold, it establishes the first step of the implication

$$q > \frac{11}{6}\Delta \implies \text{coupling independence on } [0, 1] \implies \text{zero-freeness near } [0, 1],$$

whose second step is Theorem 1.2. We obtain the first step by extending Vigoda's flip coupling [Vig00] to positive activities, conditioning on random hard constraints. At equality, we combine the positive-activity coupling estimates with the hard-colouring coupling-independence bound of [CFG⁺25, Theorem 20]; the normalization and pinning conventions are checked in Corollary 4.13. Only the equality case $q = 11\Delta/6$ of Theorem 1.1 uses this external input.

Theorem 4.1 (Hamming coupling independence on $[0, 1]$). *Let q, Δ be positive integers with $\Delta \geq 2$ and $q > 11\Delta/6$. Put*

$$\gamma_0 := \frac{q}{\Delta} - \frac{11}{6} > 0.$$

Then, for every $G \in \mathcal{G}_\Delta$, every pinning τ , every $x \in [0, 1]$, every $r \in V^\tau$, and all $a, b \in [q]$,

$$W_{1,\text{Ham}}(\mu_{G,x}^{\tau^{r=a}}, \mu_{G,x}^{\tau^{r=b}}) \leq \frac{2(1-x)\Delta}{q - (11/6)(1-x)\Delta} \leq \frac{2}{\gamma_0}.$$

At $x = 0$ both laws are well defined by Lemma 2.1, since the hypotheses imply $q \geq \Delta + 2$.

To compare the laws with two different root colours, we compare each to the common system obtained by deleting the root. Pinning the root adds at most Δ boundary constraints to this common system. We need two estimates: how much these constraints change one update, and how strongly the kernel contracts. Stationary comparison converts them into a distance between stationary laws; the triangle inequality then gives coupling independence. Section 4.1 records stationary comparison and the endpoint continuity that covers $x = 0$; Sections 4.2 and 4.3 construct the kernels and bound the boundary effect; Section 4.4 proves the conditional one-step estimate; Section 4.5 averages it and applies stationary comparison, proving Theorem 4.1 and Corollary 4.13; and Section 4.6 proves Theorem 1.1.

4.1 Stationary comparison and endpoint continuity

For a Markov kernel P , write νP for the law after one update from ν . We use the following variant of [CFG⁺25, Lemma 19], in which the contracting chain is called P , contraction is assumed only for adjacent states of a path metric, and the disagreement is averaged under ν_Q rather than bounded pointwise.

Lemma 4.2 (Stationary comparison). *Let d be the shortest-path metric of a finite connected state graph with positive edge lengths. Let P, Q be Markov kernels on the vertex set of this graph, with stationary laws ν_P, ν_Q . Suppose $0 < \kappa \leq 1$, $C \geq 0$, and*

$$W_{1,d}(P(\xi, \cdot), P(\xi', \cdot)) \leq (1 - \kappa)d(\xi, \xi') \quad \text{for adjacent } \xi, \xi', \quad \mathbb{E}_{\xi \sim \nu_Q} W_{1,d}(P(\xi, \cdot), Q(\xi, \cdot)) \leq C.$$

Then $W_{1,d}(\nu_P, \nu_Q) \leq C/\kappa$.

Proof. Summing along shortest paths extends the first bound to all states; coupling the initial states then gives $W_{1,d}(\nu P, \nu' P) \leq (1 - \kappa)W_{1,d}(\nu, \nu')$ for all laws ν, ν' . Stationarity and the second hypothesis yield

$$W_{1,d}(\nu_P, \nu_Q) \leq W_{1,d}(\nu_P P, \nu_Q P) + W_{1,d}(\nu_Q P, \nu_Q Q) \leq (1 - \kappa)W_{1,d}(\nu_P, \nu_Q) + C.$$

Rearranging gives the claim. $\square$

Lemma 4.3 (Finite-state continuity of normalized laws). *On a finite space Ω , let two families of nonnegative weights be continuous at $x = 0$, with positive total weights there. Their normalized laws $\mu_x^{(0)}, \mu_x^{(1)}$ converge in total variation to $\mu_0^{(0)}, \mu_0^{(1)}$ as $x \downarrow 0$, and for every metric d on Ω ,*

$$W_{1,d}(\mu_x^{(0)}, \mu_x^{(1)}) \longrightarrow W_{1,d}(\mu_0^{(0)}, \mu_0^{(1)}).$$

Consequently, if $W_{1,d}(\mu_x^{(0)}, \mu_x^{(1)}) \leq B(x)$ for all small $x > 0$, then $W_{1,d}(\mu_0^{(0)}, \mu_0^{(1)}) \leq \liminf_{x \downarrow 0} B(x)$.

Proof. The normalized weights converge pointwise. Use $W_{1,d}(\nu, \nu') \leq \text{diam}_d(\Omega)\|\nu - \nu'\|_{\text{TV}}$ and the triangle inequality. $\square$

4.2 Random hard constraints and the soft flip kernel

Fix $G \in \mathcal{G}_\Delta$, a pinning τ , and $x \in (0, 1]$. Let $\mathcal{E}_\tau$ consist of the free edges and the individually labelled free-pinned edges; pinned-only edges have already been removed by normalization. Each member imposes an inequality between its endpoint colours. The identity

$$x\mathbf{1}_{[c=c']} + \mathbf{1}_{[c \neq c']} = x + (1-x)\mathbf{1}_{[c \neq c']} \quad (21)$$

gives the following representation. Given σ , activate each satisfied inequality independently with probability $1-x$. For the active set A , let E_A consist of its free edges and put

$$L_{A,u} = [q] \setminus \{\tau_p : up \in A \text{ is a free-pinned edge}\}, \quad F_A = (V^\tau, E_A, L_A).$$

For any hard list instance $F = (W, E, L)$, write Ω_F for its proper colourings respecting every list.

Lemma 4.4 (Soft Potts as a random hard-colouring model). *If $\sigma \sim \mu_{G,x}^\tau$ and A is sampled as above, then σ , conditional on A , is uniform on Ω_{F_A} .*

Proof. For $M = |\mathcal{E}_\tau|$, the joint unnormalized weight is

$$x^{M-|A|}(1-x)^{|A|}\mathbf{1}_{[\sigma \in \Omega_{F_A}]}. \quad (22)$$

Summing over A gives a factor 1 for each satisfied constraint and x for each unsatisfied one, hence the Potts weight. For fixed A , the displayed weight is constant on Ω_{F_A} . $\square$

A *flip profile* is a sequence $\rho = (\rho_s)_{s \geq 1}$ with $0 \leq \rho_s \leq 1$ for all s .

Definition 4.5 (Hard list-flip transition). From $X \in \Omega_F$, choose (u, c) uniformly in $W \times [q]$. If $c = X_u$, hold. Otherwise, let $S_X(u, c)$ be the $\{X_u, c\}$ -component of u , that is, the component containing u of the subgraph of (W, E) induced by the vertices coloured X_u or c . Swap its two colours with probability $\rho_{|S_X(u, c)|}/|S_X(u, c)|$, provided the swap respects every list; otherwise hold. Denote this kernel by Φ_F^ρ , taking the identity kernel when $W = \emptyset$.

A feasible component swap of size s has s proposals, hence total probability $\rho_s/(|W|q)$. The swap is an involution with the same reverse rate, so Φ_F^ρ is reversible for the uniform law.

Definition 4.6 (Soft flip kernel). The kernel $K_{x,\rho}^{G,\tau}$ samples A conditional on the current colouring, applies $\Phi_{F_A}^\rho$ (Definition 4.5), and returns the updated colouring to G , keeping τ fixed (Figure 1).

Lemma 4.7 (Soft flip kernels). *For every profile ρ and $x \in (0, 1]$, $K_{x,\rho}^{G,\tau}$ is reversible with stationary law $\mu_{G,x}^\tau$. If $\rho_1 > 0$, it is irreducible on $[q]^{V^\tau}$.*

Proof. By Lemma 4.4, conditional detailed balance for each A , multiplied by its marginal probability in (22) and summed over A , gives detailed balance for the spin marginal. The empty active set has positive probability when $x > 0$; on this event, $\rho_1 > 0$ permits every one-site recolouring. $\square$

4.3 Common activations and boundary sensitivity

For colourings X, Y at Hamming distance one (adjacent colourings), differing at a vertex v , which we call the *root*, with $X_v = a \neq b = Y_v$, draw one independent Bernoulli($1-x$) coin per labelled constraint. Activate that constraint on either side exactly when the coin is one and the inequality is satisfied there. Write $\mathcal{A} = (A_X, A_Y)$ for the outcome and F_X, F_Y for the hard instances.

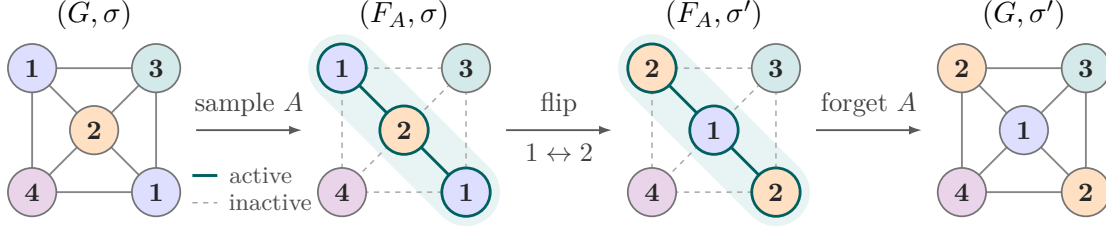


Figure 1: One soft flip update (Definition 4.6), without pins. Given σ , each satisfied edge is activated independently with probability $1 - x$; inactive (dashed) edges are absent from the hard instance F_A . The flip kernel swaps the highlighted $\{1, 2\}$ -component of F_A , and discarding A returns the updated colouring σ' to G .

Lemma 4.8 (The two hard instances differ only near v). *The common-coin construction has the correct activation marginals. The two instances have the same induced graph and lists off v . Their sets of edges at v can differ only at neighbours of colour a or b , and their lists at v can differ only in those two colours.*

Proof. An inequality has different satisfaction status only if it is incident to v and its other endpoint has colour a or b . $\square$

Lemma 4.9 (Effect of adding one labelled free-pinned edge). *Let a minus and a plus pinned system have the same free vertex set W , of size $m \geq 1$, and the same labelled constraints except for one additional free-pinned edge in the plus system. For a profile ρ with $R_\rho := \sup_{s \geq 1} s\rho_s < \infty$ and $x \in (0, 1]$, their soft kernels K_- and K_+ satisfy*

$$W_{1, \text{Ham}}(K_-(\sigma, \cdot), K_+(\sigma, \cdot)) \leq \frac{(1-x)R_\rho}{mq} \quad (\sigma \in [q]^W). \quad (23)$$

For hard list kernels, deleting one colour from one list gives the bound $R_\rho/(mq)$ on colourings feasible for the smaller list. Adding k labelled free-pinned edges (for hard list kernels, deleting k colours from lists) multiplies the respective bound by k .

Proof. Couple the common activation coins and flip proposals. Suppose the extra constraint is the edge up , which forbids the colour τ_p at u . It is inactive when $\sigma_u = \tau_p$; otherwise it is active with probability $1 - x$. When active, it can change acceptance only for the unique $\{\sigma_u, \tau_p\}$ -component containing u . If this component has size s , couple its rejected plus move to its minus move: the cost is at most $s\rho_s/(mq) \leq R_\rho/(mq)$. Average over activation to prove (23); deterministic list deletion omits the factor $1 - x$. For several free-pinned edges (or deleted colours), add them one at a time and use the triangle inequality. $\square$

4.4 The conditional hard estimate

Use Vigoda's profile [Vig00]

$$p_1 = 1, \quad p_2 = \frac{13}{42}, \quad p_3 = \frac{1}{6}, \quad p_4 = \frac{2}{21}, \quad p_5 = \frac{1}{21}, \quad p_6 = \frac{1}{84}, \quad p_s = 0 \quad (s \geq 7), \quad (24)$$

and abbreviate $\Phi_F := \Phi_F^{\mathbf{p}}$ and $K_x^{G, \tau} := K_{x, \mathbf{p}}^{G, \tau}$.

Fix $G \in \mathcal{G}_\Delta$, a pinning τ , and $x \in (0, 1]$. For colourings X, Y at Hamming distance one, with $X_v = a \neq b = Y_v$, condition on their common-coin activation outcome $\mathcal{A}$. Write F_X, F_Y for the

resulting hard instances, $L_{X,v} := L_{A_X,v}$ and $L_{Y,v} := L_{A_Y,v}$ for their root lists, $n = |V^\tau|$, and G_0 for their common induced graph off v , with its common colouring. The two current colourings are feasible, so $a \in L_{X,v}$ and $b \in L_{Y,v}$; the root lists agree on every other colour. Put

$$N_c = \{u \in N_{G^\tau}(v) : X_u = Y_u = c, vu \text{ is active on at least one side}\}, \quad \ell = |L_{X,v} \cap L_{Y,v}|.$$

The one-step estimate of Vigoda [Vig00] and its list-colouring extension in [CDM⁺19, Section 6 and Appendix E] couple two colourings of one fixed instance that differ at a single vertex. Here the two chains run on the hard instances F_X and F_Y , which may differ in the edges at the root and in the root list (Lemma 4.8), so we prove the estimate directly.

Proposition 4.10 (Conditional hard estimate). *The updates admit a coupling (X', Y') , with $X' \sim \Phi_{F_X}(X, \cdot)$ and $Y' \sim \Phi_{F_Y}(Y, \cdot)$, such that*

$$nq \mathbb{E}[\text{Ham}(X', Y') - 1 \mid \mathcal{A}] \leq \frac{11}{6} \sum_c |N_c| - \ell. \quad (25)$$

Proof. We first match component moves and then bound the Hamming drift separately for each colour. Detailed size estimates are needed only when $c \notin \{a, b\}$ and $|N_c| \in \{1, 2\}$; the other cases admit coarse bounds.

Scale all move probabilities by nq . Write $p(M)$ for the scaled mass of a component move M : it is $p_{|M|}$ if the swap is feasible and zero otherwise. A root component contains v ; an off-root component does not. Match every move that is off-root in both instances with the identical move in the other instance.

Call $c \notin \{a, b\}$ a regular colour. Every active edge from v to N_c is then common to both instances. Let $\mathcal{C}$ be the family of $\{a, c\}$ -components of G_0 meeting N_c , and let $\mathcal{D}$ be their $\{b, c\}$ -counterparts. Adding v joins each family into a root component on one side; on the other side its members remain off-root. The remaining moves, which form group c , are:

colour pair	moves in X	moves in Y
$\{a, c\}$	root R	off-root $C \in \mathcal{C}$
$\{b, c\}$	off-root $D \in \mathcal{D}$	root S

Here

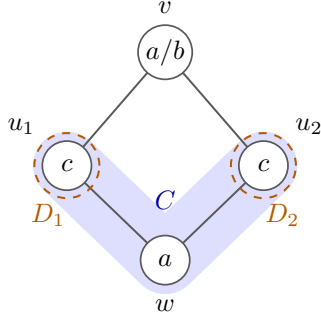
$$R = \{v\} \cup \bigcup_{C \in \mathcal{C}} C, \quad S = \{v\} \cup \bigcup_{D \in \mathcal{D}} D.$$

A root swap is feasible exactly when its new root colour is allowed and all its constituent off-root swaps are feasible: these are precisely its vertexwise list tests. This equivalence will also handle blocked moves in the estimates below.

Matching the moves. Write $m = |N_c|$. If $m = 0$, match the two singleton root moves, of mass one, when c is in both root lists; otherwise both are blocked. If $m \geq 1$, choose largest components $C_\star, D_\star$, breaking ties by the least-indexed neighbour in a fixed ordering of N_c . Match the whole mass $p(R)$ of R to $C_\star$, and the whole mass $p(S)$ of S to $D_\star$. These allocations fit: since $|R| > |C_\star|$ and p_s is nonincreasing for $s \geq 1$, we have $p_{|R|} \leq p_{|C_\star|}$, and if the swap of R is feasible, so is the swap of its constituent $C_\star$, by the equivalence above. Hence $p(R) \leq p(C_\star)$, and likewise $p(S) \leq p(D_\star)$. The residual off-root masses are

$$\tilde{p}(C) = p(C) - \mathbf{1}_{[C=C_\star]}p(R), \quad \tilde{p}(D) = p(D) - \mathbf{1}_{[D=D_\star]}p(S).$$

Shared off-root component



Root matches and incidence allocation

$$R_X \longleftrightarrow C_Y : 8, \quad (D_1)_X \longleftrightarrow S_Y : 14.$$

Move	Initial	After root matching	Used at u_1	Used at u_2	Left
C_Y	14	6	3	3	0
$(D_1)_X$	84	70	3	0	67
$(D_2)_X$	84	84	0	3	81

$$u_1 : \min\{6/2, 70/1\} = 3, \quad u_2 : \min\{6/2, 84/1\} = 3.$$

Figure 2: Incidence matching on a four-cycle with full lists, $q = 4$, and all four edges active. The root has colour a in X and b in Y . Off the root, $C = \{u_1, w, u_2\}$ and $D_i = \{u_i\}$, so $|R| = 4$ and $|S| = 3$; choose $D_\star = D_1$. Subscripts X, Y indicate the instance in which a move is made. All numbers are 84 times the scaled masses and hence represent actual probabilities in units of $1/(84nq)$. By (24), $84p_4 = 8$ for R , $84p_3 = 14$ for C and S , and $84p_1 = 84$ for D_1 and D_2 . The root moves R_X and S_Y are matched in full, so they leave no residual and have no row. Each of the two “Used at” columns records both sides of the same match. The two incidences use $3 + 3 = 6$, exactly the residual mass of C ; without the division by $|C \cap N_c| = 2$, each would claim 6, twice what C has left.

Write $N_c = \{u_1, \dots, u_m\}$ in the fixed ordering. For each i , let $C_i \in \mathcal{C}$ and $D_i \in \mathcal{D}$ be the components containing u_i , and match an additional mass

$$\min \left\{ \frac{\tilde{p}(C_i)}{|C_i \cap N_c|}, \frac{\tilde{p}(D_i)}{|D_i \cap N_c|} \right\} \quad (26)$$

between those two off-root moves. Dividing by the number of incidences ensures that no component’s residual mass is exceeded. Figure 2 illustrates this division when a single component meets two root neighbours.

For the pair $\{a, b\}$, the Y -root consists of v and the $\{a, b\}$ -components of G_0 meeting N_a ; the X -root consists of v and those meeting N_b . A component meeting both sets belongs to both roots and gives no off-root move. Put the Y -root move, together with the X -off-root moves meeting N_a but not N_b , into group a , and define group b symmetrically. When $|N_a| = 1$ and its component C is off-root on the X -side, match the whole Y -root mass to C . Otherwise match it to holding. Do the symmetric construction for b . The match with C fits: the Y -root is then $\{v\} \cup C$, so its mass is at most $p_{|C|+1} \leq p_{|C|}$, and if the Y -root swap is feasible, so is the swap of C in X , since the vertices of C have the same lists in both instances (Lemma 4.8). Each side has at least n units of scaled holding mass, contributed by the n proposals in which a vertex proposes its current colour. On each side, at most one root move uses the holding mass, and its scaled mass is at most $p_1 = 1$, so these allocations fit.

These groups exhaust the remaining side-labelled moves without overlap: colour pairs avoiding a, b were common, and the preceding description classifies all pairs containing either root colour. Complete the partial coupling by pairing the residual masses of X -moves and Y -moves proportionally: if they are ω_M and $\omega'_{M'}$, with common total $T > 0$, add mass $\omega_M \omega'_{M'} / T$ to the pair (M, M') ; if $T = 0$, nothing remains. All allocations have the required marginals after division by nq .

The charges. A pair of moves M, M' coupled by this completion increases the Hamming distance by at most $|M| + |M'|$. Charge each residual move its size times its residual mass; let Q_c be

these charges plus the actual drift of the prescribed matches in group c . Common matches have zero drift, and the residual cost separates by side even when the completion joins different groups. Consequently, the first bound below holds, and it suffices to prove the second for every colour c :

$$nq \mathbb{E}[\text{Ham}(X', Y') - 1 \mid \mathcal{A}] \leq \sum_c Q_c, \quad (27)$$

$$Q_c \leq \frac{11}{6} |N_c| - \mathbf{1}_{[c \in L_{X,v} \cap L_{Y,v}]}. \quad (28)$$

For a regular colour, matching R to $C_\star$ creates disagreements only on the other constituents of R ; its drift is $|R| - |C_\star| - 1$. The analogous cost for S is $|S| - |D_\star| - 1$. Each off-root match in (26) saves at least one unit because $C_i \cap D_i$ contains u_i . Adding root-match costs and residual size charges, and subtracting these matching savings, gives, for $m \geq 1$,

$$\begin{aligned} Q_c &\leq (|R| - |C_\star| - 1)p(R) + (|S| - |D_\star| - 1)p(S) \\ &\quad + \sum_C |C| \tilde{p}(C) + \sum_D |D| \tilde{p}(D) - \sum_{i=1}^m \min \left\{ \frac{\tilde{p}(C_i)}{|C_i \cap N_c|}, \frac{\tilde{p}(D_i)}{|D_i \cap N_c|} \right\}. \end{aligned} \quad (29)$$

The following bounds follow directly from the six nonzero profile values:

$$rp_r \leq 1, \quad rp_r \leq \frac{13}{21} \ (r \geq 2), \quad (r-1)p_r \leq \frac{1}{3}, \quad (r-2)p_r \leq \frac{4}{21}. \quad (30)$$

For two off-root moves meeting the same neighbour, with sizes r, s and masses θ, θ' , write $g_{r,s}(\theta, \theta') = r\theta + s\theta' - \min\{\theta, \theta'\}$ for their size charges minus the guaranteed matching saving. This function is nondecreasing in each argument for $r, s \geq 1$. Since p_r is nonincreasing, $g_{r,s}(p_r, p_s) = rp_r + (s-1)p_s$ when $r \leq s$. Using (30), with $r = 1$ or $r \geq 2$, gives

$$g_{r,s}(\theta/j, \theta'/k) \leq \frac{4}{3} \quad (0 \leq \theta \leq p_r, \ 0 \leq \theta' \leq p_s, \ j, k \geq 1). \quad (31)$$

Splitting each component charge among its incidences in N_c now bounds the last three terms of (29) by $4m/3$. If c is blocked at the root, both root masses vanish, proving (28). If it is allowed and $m \geq 3$, the root terms are each at most $4/21$, giving $Q_c \leq 8/21 + 4m/3 \leq 11m/6 - 1$. For $m = 0$, the singleton match gives $Q_c = -\mathbf{1}_{[c \in L_{X,v} \cap L_{Y,v}]}$.

Regular colours with one or two neighbours. It remains to treat an allowed regular colour with $m \in \{1, 2\}$. For $m = 1$, let the two off-root sizes be r, s and put $d_r = p_r - p_{r+1}$. Root feasibility equals off-root feasibility, so the residual masses lie in $\{0, d_r\}$ and $\{0, d_s\}$, respectively. Thus (29) and monotonicity of g give $Q_c \leq g_{r,s}(d_r, d_s)$. The profile gives

$$d_1 = \frac{29}{42}, \quad rd_r \leq \frac{2}{7} \ (r \geq 2), \quad (r-1)d_r \leq \frac{1}{7} \ (r \geq 2).$$

If both sizes are at least two, the bound is at most $4/7$. If both are one, it is $29/42$; otherwise it is at most $29/42 + 1/7 = 5/6$, as required.

For $m = 2$, first suppose that some off-root component contains both neighbours. Its size is at least three, since those neighbours have the same colour. Its root term in (29) is zero, and its off-root charge is at most $rp_r \leq 1/2$ for $r \geq 3$. The other family contributes at most two in off-root charges. Its root term is zero for one component, and at most $k'p_{1+k+k'} \leq 1/6$ for two sizes $k \geq k'$. Hence $Q_c \leq 1/2 + 2 + 1/6 = 8/3$.

Otherwise each family has two distinct components. Index them by their neighbours, with sizes r_1, r_2 and s_1, s_2 , and feasibility bits $f_i, f'_i \in \{0, 1\}$. Let $P = p(R)$ and $P' = p(S)$. Before allocating mass to the root matches, the off-root bound is

$$H_0 = \sum_{i=1}^2 g_{r_i, s_i}(f_i p_{r_i}, f'_i p_{s_i}).$$

For $\mathcal{C}$, the root match costs $(\min_i r_i)P$, removes size charge $(\max_i r_i)P$, and loses at most P of matching saving. These three contributions explain its correction to H_0 . Adding the corresponding correction for $\mathcal{D}$ gives

$$Q_c \leq H_0 + (1 - \max r_i + \min r_i)P + (1 - \max s_i + \min s_i)P'. \quad (32)$$

A correction is nonpositive for unequal sizes. For two equal sizes $k \geq 2$, it is at most p_{2k+1} , which is nonzero only for $k = 2$ and then equals $1/21$. For two singletons it is at most $p_3 = 1/6$.

Thus positive corrections occur only within equal-size families. Singletons also have the largest off-root mass, whereas $rp_r \leq 13/21$ for every nonsingleton. This suggests classifying the four sizes by their number h of singletons. A pair of singletons has $g_{1,1} \leq 1$; a pair of nonsingletons has $g_{r,s} \leq rp_r + sp_s \leq 26/21$ by (30); and every pair satisfies $g_{r,s} \leq 4/3$ by (31). These bounds give the following upper bounds in (32):

h	arrangement	H_0	total correction	Q_c
0	any	$52/21$	$2/21$	$18/7$
1	any	$4/3 + 26/21$	$1/21$	$55/21$
2	one singleton in each family	$2 \cdot 4/3$	0	$8/3$
3	any	$1 + 4/3$	$1/6$	$5/2$
4	any	2	$2 \cdot 1/6$	$7/3$

Every last-column entry is at most $8/3$.

It remains to handle $h = 2$ with both singletons in one family, say $\mathcal{C}$. Here the general correction bound is too coarse. If $P > 0$, both singleton swaps are feasible and their residual masses are at least $1 - p_3 = 5/6 > p_2$. Since both opposite masses are at most p_2 , subtracting P loses no matching saving. The root cost P and removed size charge P cancel, so $\mathcal{C}$ has zero correction. The $\mathcal{D}$ -correction is nonpositive for unequal sizes and zero for equal sizes at least three. Only $(s_1, s_2) = (2, 2)$ remains, where

$$Q_c \leq 2g_{1,2}(p_1, p_2) + p_5 = 2(1 + p_2) + p_5 = \frac{8}{3};$$

otherwise $H_0 \leq 8/3$ suffices. This proves the regular-colour bound entirely by the displayed scalar inequalities.

The two root colours. Consider colour a , writing $m = |N_a|$; colour b is symmetric. If $a \notin L_{Y,v}$, its root move is blocked and its at most m off-root components each cost at most $sp_s \leq 1$. If $a \in L_{Y,v}$ and $m = 0$, its singleton root/holding match coalesces, giving $Q_a = -1$. For $m \geq 2$, let r be the size of the Y -root. Its root/holding match has drift $r - 2$ and mass at most p_r , contributing at most $(r - 2)p_r \leq 4/21$; the at most m off-root components again cost at most one each, so $Q_a \leq m + 4/21 \leq 11m/6 - 1$.

In the remaining case $a \in L_{Y,v}$ and $m = 1$, its root is $\{v\} \cup C$, where C has size s . If the root swap is infeasible, the corresponding off-root swap, when present, fails the same list test, giving zero

charge. If the root swap is feasible and C also meets N_b , there is no off-root move in this group and the charge is at most $(s-1)p_{s+1} \leq 4/21$. Otherwise the root/off-root match coalesces, leaving charge at most

$$-p_{s+1} + s(p_s - p_{s+1}) = sp_s - (s+1)p_{s+1} \leq \frac{8}{21} < \frac{5}{6}.$$

This proves (28) for all colours; summing proves the proposition. $\square$

Arithmetic check. The script `verifier.py`, attached to this PDF and available as an ancillary file with the arXiv source, independently checks the bounds $rp_r \leq 1$ and $(r-2)p_r \leq 4/21$, (31), the two one-neighbour root-colour bounds, and the maximum of the right-hand side of (29) for $m \leq 2$. Run it with `python3 verifier.py`. Using exact rational arithmetic, it enumerates component partitions, component sizes capped at seven, and off-root feasibility bits, deriving root feasibility from the list-test equivalence above; the analytic proof itself needs no enumeration.

4.5 Averaging and stationary comparison

Lemma 4.11 (Soft path contraction). *For $G \in \mathcal{G}_\Delta$, a pinning τ , and $x \in (0, 1]$, put $t = 1 - x$ and $n = |V^\tau| \geq 1$. Colourings X, Y at Hamming distance one admit a coupling (X', Y') of one step of $K_x^{G, \tau}$ satisfying*

$$nq \mathbb{E}[\text{Ham}(X', Y') - 1] \leq \frac{11}{6}t\Delta - q. \quad (33)$$

In particular, if $q > (11/6)t\Delta$, the soft kernel contracts Hamming distance by the factor $1 - \kappa_x/n$, where

$$\kappa_x = \frac{q - (11/6)t\Delta}{q} > 0. \quad (34)$$

Proof. At the disagreement vertex v , let $d_v := \deg_{G^\tau}(v)$ be its free degree, $d_\Lambda := |N_G(v) \cap \Lambda|$ its number of pinned neighbours, and k the number of free–pinned edges at v that are active on at least one side. Each constraint at v is satisfied on at least one side, so it is active on at least one side exactly when its Bernoulli(t) coin is one. Thus

$$\mathbb{E} \sum_c |N_c| = td_v, \quad \mathbb{E} k = td_\Lambda, \quad \ell \geq q - k.$$

Averaging Proposition 4.10 gives, since $d_v + d_\Lambda = \deg_G(v) \leq \Delta$,

$$nq \mathbb{E}[\text{Ham}(X', Y') - 1] \leq \frac{11}{6}td_v + td_\Lambda - q \leq \frac{11}{6}t\Delta - q.$$

Divide by nq ; path coupling [BD97] along shortest paths extends the adjacent-state bound to all pairs. $\square$

Proposition 4.12 (Vigoda-profile coupling for newly pinned laws). *Let q, Δ be positive integers with $\Delta \geq 2$. For every $G \in \mathcal{G}_\Delta$, every pinning τ , every $r \in V^\tau$, all $a, b \in [q]$, and every $x \in (0, 1]$ satisfying $q > (11/6)(1-x)\Delta$,*

$$W_{1, \text{Ham}}(\mu_{G, x}^{\tau^{r=a}}, \mu_{G, x}^{\tau^{r=b}}) \leq \frac{2(1-x)\Delta}{q - (11/6)(1-x)\Delta}. \quad (35)$$

If $q > 11\Delta/6$, the bound (35) also holds at $x = 0$.

Proof. The cases $a = b$ or $m := |V^\tau \setminus \{r\}| = 0$ are immediate. Otherwise compare each newly pinned system to the common middle system $(G - r, \tau)$, whose Gibbs law $\mu_{\text{mid}} := \mu_{G-r, x}^\tau$ is stationary for $K_x^{G-r, \tau}$ (Lemma 4.7). The child of colour c adds at most Δ labelled free-pinned edges to this middle system. Since $\sup_s sp_s = 1$, Lemma 4.9 gives, for every common state σ ,

$$W_{1, \text{Ham}}(K_x^{G, \tau^{r=c}}(\sigma, \cdot), K_x^{G-r, \tau}(\sigma, \cdot)) \leq \frac{(1-x)\Delta}{mq}. \quad (36)$$

By Lemma 4.11, applied to the child system with its m free vertices, the child kernel contracts by the factor $1 - \kappa_x/m$. Apply Lemma 4.2 with $\kappa = \kappa_x/m$ and $C = (1-x)\Delta/(mq)$, using the stationary laws from Lemma 4.7, and then the triangle inequality:

$$W_{1, \text{Ham}}(\mu_{G, x}^{\tau^{r=c}}, \mu_{\text{mid}}) \leq \frac{(1-x)\Delta}{q\kappa_x}, \quad W_{1, \text{Ham}}(\mu_{G, x}^{\tau^{r=a}}, \mu_{G, x}^{\tau^{r=b}}) \leq \frac{2(1-x)\Delta}{q\kappa_x}.$$

This is (35). If $q > 11\Delta/6$, both hard pinned laws are well defined, with positive total weights, by Lemma 2.1; their weights are polynomial in x and the right-hand side of (35) is continuous at $x = 0$, so Lemma 4.3 passes the bound to $x = 0$. The right-hand side of (35) is largest at $x = 0$, where it equals $2/\gamma_0$; this proves Theorem 4.1. $\square$

Corollary 4.13 (Coupling input at $q = 11\Delta/6$). *Let q, Δ be integers with $\Delta \geq 2$ and $q = 11\Delta/6$. Then $\mathcal{G}_\Delta$ satisfies coupling independence at $x = 0$ with a finite constant, and for every $\delta \in (0, 1]$ it satisfies coupling independence on $[\delta, 1]$ with constant $12/(11\delta)$.*

Proof. For $x \in [\delta, 1]$, Proposition 4.12 gives

$$W_{1, \text{Ham}}(\mu_{G, x}^{\tau^{r=a}}, \mu_{G, x}^{\tau^{r=b}}) \leq \frac{12(1-x)}{11x} \leq \frac{12}{11\delta}.$$

At $x = 0$, integrality forces Δ to be a positive multiple of 6, so [CFG⁺25, Theorem 20 and Proposition 22] apply. The conditional weights in [CFG⁺25, Section 2] omit pinned-only edges and allow improper pinnings, agreeing with our normalization. Apply the coupling bound of [CFG⁺25, Theorem 20] to $\tau^{r=a}$ and $\tau^{r=b}$, then project onto $V^\tau \setminus \{r\}$, to obtain the bound required by Definition 2.2. As noted in [CFG⁺25], this hard-colouring bound is implicit in the coupling arguments of [Liu21, BCC⁺22]. Its list-contraction input, [CFG⁺25, Proposition 22], is derived there from the list-colouring analysis of [CDM⁺19, Section 6 and Appendix E], which treats lists of equal size; we use [CFG⁺25, Theorem 20] as stated. $\square$

4.6 Proof of the Potts zero-free theorem

Proof of Theorem 1.1. The hypotheses imply $q \geq \Delta + 2$. If $q > 11\Delta/6$, then Theorem 4.1 gives one coupling-independence constant on all of $[0, 1]$. If $q = 11\Delta/6$, then Corollary 4.13 supplies the two coupling inputs required by Theorem 1.2. Applying Theorem 1.2 to $\mathcal{G}_\Delta$ gives the normalized zero-free statement. The assertion for the unnormalized pinned polynomial follows from (3). $\square$

5 Zero-freeness for colour fields and Holant problems

This section applies the separator induction of Section 3 to two further families. Section 5.1 perturbs the colour fields of proper colourings around the uniform field; only coupling independence at $x = 0$ is needed. Section 5.2 treats log-concave Boolean Holant problems, where pinned instances carry residual signatures and the coupling input comes from [CG24].

5.1 Colour fields

5.1.1 Vertex-colour fields

In the spirit of the Lee–Yang theorem [LY52], we now consider zeros in complex colour fields. We use the same letter Z for both partition functions: an argument z denotes the edge-activity polynomial, while an argument λ denotes the field polynomial defined below. For $G = (V, E) \in \mathcal{G}_\Delta$, a pinning $\tau: \Lambda \rightarrow [q]$, and field variables $\lambda = (\lambda_{u,c})_{(u,c) \in V \times [q]}$, define the normalized pinned field polynomial

$$\hat{Z}_G^\tau(\lambda) := \sum_{\varphi: V^\tau \rightarrow [q]} \prod_{u \in V^\tau} \lambda_{u, \varphi_u} \prod_{\substack{uv \in E \\ \{u,v\} \not\subseteq \Lambda}} \mathbf{1}_{[(\tau \cup \varphi)_u \neq (\tau \cup \varphi)_v]}. \quad (37)$$

At the all-ones vector $\lambda = \mathbf{1}$ on $V \times [q]$, this equals the hard-colouring value $\hat{Z}_G^\tau(z)|_{z=0}$ of (5), which is positive when $q \geq \Delta + 1$ (Lemma 2.1). The unpinned field polynomial is $Z_G(\lambda) = \hat{Z}_G^\emptyset(\lambda)$. The pinned field polynomial is

$$Z_G^\tau(\lambda) := \sum_{\substack{\sigma: V \rightarrow [q] \text{ proper} \\ \sigma|_\Lambda = \tau}} \prod_{v \in V} \lambda_{v, \sigma_v}.$$

If τ is proper on $G[\Lambda]$, then

$$Z_G^\tau(\lambda) = \left(\prod_{v \in \Lambda} \lambda_{v, \tau_v} \right) \hat{Z}_G^\tau(\lambda|_{V^\tau \times [q]}); \quad (38)$$

otherwise $Z_G^\tau(\lambda) \equiv 0$.

For $v \in V^\tau$, conditioning on its colour gives

$$\hat{Z}_G^\tau(\lambda) = \sum_{a: b_v^\tau(a)=0} \lambda_{v,a} \hat{Z}_G^{\tau^{v=a}}(\lambda|_{(V^\tau \setminus \{v\}) \times [q]}). \quad (39)$$

The normalized child $\hat{Z}_G^{\tau^{v=a}}$ remains defined when a is blocked at v ; such a colour simply does not occur in (39).

The following transfer needs coupling independence only at $x = 0$.

Proposition 5.1 (Field transfer). *Fix integers q, Δ with $\Delta \geq 2$ and $q \geq \Delta + 1$, let $C_0 \in [0, \infty)$, and let $\mathcal{G} \subseteq \mathcal{G}_\Delta$ be closed under taking induced subgraphs and satisfy C_0 -coupling independence at $x = 0$. There is $\theta = \theta(q, \Delta, C_0) \in (0, 1/2]$ such that, for every $G \in \mathcal{G}$ and every pinning τ ,*

$$\hat{Z}_G^\tau(\lambda) \neq 0 \quad \text{whenever } |\lambda_{u,c} - 1| < \theta \quad (u \in V^\tau, c \in [q]).$$

Proof. We use the separator induction of Lemma 3.2, with base field $\mathbf{1}$ and coupling constant C_0 . Choose L, s, N, α as in that proof. Up to relabelling, only finitely many local polynomials occur below (bounded components and the interior polynomials $D_{c,\xi}$), each in at most $(N + s)q$ field variables; fix $\theta \in (0, 1/2]$ so small that the bounds on them used below hold on the polydisc $\max_{u,c} |\lambda_{u,c} - 1| < \theta$. Induct on the number of free vertices, proving nonvanishing and the response bound

$$\left| \text{Log} \frac{\hat{Z}_G^{\tau^{v=a}}(\lambda) / \hat{Z}_G^{\tau^{v=a}}(\mathbf{1})}{\hat{Z}_G^{\tau^{v=b}}(\lambda) / \hat{Z}_G^{\tau^{v=b}}(\mathbf{1})} \right| \leq \alpha. \quad (40)$$

All normalized pinned base values are positive by Lemma 2.1, and logarithms continue from $\lambda = \mathbf{1}$ along the segment to λ , which stays in the polydisc. Bounded components are handled by continuity over finitely many local instances.

For a larger component, choose a shell S with $W_{1,\text{Ham}}(\nu_a, \nu_b) \leq C_0/L$. Include the interior and shell field weights in the local polynomial $D_{c,\xi}(\lambda)$, so that

$$\widehat{Z}_G^{\tau^{v=c}}(\lambda) = \sum_{\xi} D_{c,\xi}(\lambda) \widehat{Z}_{G'}^{\tau_{\xi}}(\lambda).$$

Field perturbations preserve the hard constraints. Thus each local polynomial is either identically zero or has positive value at $\mathbf{1}$. By the choice of θ , every nonzero local polynomial satisfies

$$\eta_c(\xi) := \text{Log} \frac{D_{c,\xi}(\lambda)}{D_{c,\xi}(\mathbf{1})}, \quad |\eta_c(\xi)| \leq \alpha/8.$$

Set $\eta_c(\xi) = 0$ when $D_{c,\xi} \equiv 0$; such shell assignments ξ have zero ν_c -mass. The exterior response

$$h_{\xi} := \text{Log} \frac{\widehat{Z}_{G'}^{\tau_{\xi}}(\lambda)}{\widehat{Z}_{G'}^{\tau_{\xi}}(\mathbf{1})}$$

is α -Lipschitz in Hamming distance by induction, applied as in the proof of Lemma 3.2 to the instance obtained by unpinning one shell vertex, and has oscillation at most $s\alpha \leq 1/16$. The separator identity becomes

$$R_c := \frac{\widehat{Z}_G^{\tau^{v=c}}(\lambda)}{\widehat{Z}_G^{\tau^{v=c}}(\mathbf{1})} = \mathbb{E}_{\nu_c}[e^{h+\eta_c}].$$

The hypotheses of Lemma 3.1, with $A = \alpha$ and $\beta = \alpha/8$, hold along the segment from $\mathbf{1}$ to λ , so, as in the proof of Lemma 3.2, the branch in Lemma 3.1 is the logarithm continued from $\mathbf{1}$. The lemma gives $|\text{Log}(R_a/R_b)| \leq 2\alpha C_0/L + \alpha/2 < \alpha$, proving (40).

Finally, take a colour $a_0 \in L^{\tau}(v)$, which exists by Lemma 2.1. After division by R_{a_0} , the parent recursion is

$$\frac{\widehat{Z}_G^{\tau}(\lambda)}{R_{a_0}} = \sum_{a: b_v^{\tau}(a)=0} \widehat{Z}_G^{\tau^{v=a}}(\mathbf{1}) \lambda_{v,a} \frac{R_a}{R_{a_0}}.$$

Each summand has positive real part: its first factor is positive, $|\arg \lambda_{v,a}| < \pi/6$ because $|\lambda_{v,a} - 1| < 1/2$, and $|\Im \text{Log}(R_a/R_{a_0})| \leq \alpha$, so its argument has absolute value less than $\pi/6 + \alpha < \pi/2$. This closes the induction. $\square$

Theorem 5.2 (Lee–Yang polydisc). *Let q, Δ be integers with $\Delta \geq 2$. Each of the following parameter assumptions yields the conclusion below:*

- (i) $q \geq (11/6 - 1/84000)\Delta$;
- (ii) $\Delta \geq 125$ and $q \geq 1.809\Delta$;
- (iii) $\Delta \geq 3$ and $q \geq \Delta + 3$.

In cases (i) and (ii), put $\mathcal{G} = \mathcal{G}_{\Delta}$. In case (iii), there is an integer $g_{\text{LY}} = g_{\text{LY}}(q, \Delta) \geq 3$ for which we put $\mathcal{G} = \mathcal{G}_{\Delta, g_{\text{LY}}}$. In each case there is $\theta = \theta(q, \Delta) \in (0, 1/2]$ such that, for every $G \in \mathcal{G}$ and every pinning τ ,

$$\widehat{Z}_G^{\tau}(\lambda) \neq 0 \quad \text{whenever } |\lambda_{u,c} - 1| < \theta \quad (u \in V^{\tau}, c \in [q]).$$

For a proper pinning, the corresponding statement for $Z_G^{\tau}(\lambda)$ follows from (38) provided $\lambda_{v,\tau_v} \neq 0$ for every $v \in \Lambda$. In particular, it holds when the same polydisc condition is imposed at every vertex and colour.

Proof. Each case implies $q \geq \Delta + 1$, and in each case $\mathcal{G}$ is closed under taking induced subgraphs. By Proposition 5.1, it suffices to show that $\mathcal{G}$ satisfies coupling independence at $x = 0$ with a finite constant $C_0 = C_0(q, \Delta)$. At $x = 0$, the two laws in Definition 2.2 are uniform distributions on list colourings: in the instance on G^τ in which r has list $[q]$ and every other free vertex u has list $L^\tau(u)$, pin r to a and to b . This is the convention of [CFG⁺25, Section 2], which allows improper pinnings (see the proof of Corollary 4.13). As in the proof of Lemma 2.1, every vertex u of this instance has a list of size at least $\deg_{G^\tau}(u) + q - \Delta$; in particular every colour of $[q]$ is feasible at r .

In case (ii), the input is [CFG⁺25, Theorem 20], based on the flip contraction of [CV25]; projecting its coupling onto $V^\tau \setminus \{r\}$ gives the bound (8) at $x = 0$. Case (i) reduces to the other inputs by integrality. If $2 \leq \Delta < 125$, then $6q \geq 11\Delta - \Delta/14000 > 11\Delta - 1$; since $6q$ and 11Δ are integers, this implies $q \geq 11\Delta/6$. Apply Theorem 4.1 at $x = 0$ when the inequality is strict, and Corollary 4.13 when equality holds. If $\Delta \geq 125$, then $q \geq (11/6 - 1/84000)\Delta > 1.809\Delta$, so case (ii) applies. The constant $1/84000$ is the value of ε_0 fixed in the proof of [CDM⁺19, Theorem 5.1].

In case (iii), [CLMM23, Theorem 5.10 and Lemmas 5.13 and 5.14] provide an integer $g = g(q, \Delta)$ and an influence-decay radius $R = R(q, \Delta)$ with the following property: for list colourings of a graph of maximum degree at most Δ and girth at least g , with lists of size at most q and at least the degree plus 3, pinning one vertex to two different colours of its list changes the law of the remaining vertices by at most $2\Delta^R$ in the distance $W_{1,\text{Ham}}$. This is recorded as coupling independence in [CFG⁺25, Section 4]. Put $g_{\text{LY}} := \max\{g, 3\}$. For $G \in \mathcal{G}_{\Delta, g_{\text{LY}}}$, the graph G^τ has maximum degree at most Δ and girth at least g , and, since $q - \Delta \geq 3$, the instance above satisfies these hypotheses. Hence $C_0 := 2\Delta^R$ works. $\square$

Remark 5.3 (Related results). For colour-class fields $\lambda_{u,c} = \lambda_c$, Kuchukova, Perkins, and Povill showed that $Z_G(\lambda) \neq 0$ whenever $q \geq 2\Delta$ and $\max_c |\lambda_c - 1|$ is at most an explicit radius of order Δ^{-4} [KPP26, Theorem 1.14 and Corollary 1.15]. Theorem 5.2 allows vertex-dependent fields and applies below 2Δ , but its radius is not explicit. For hypergraph colourings in the local lemma regime, Liu and Yu obtained a Lee–Yang zero-free strip around $[0, 1]$ [LY26].

5.1.2 Edge-colour fields

For a finite simple graph $G = (V, E)$, let $L(G)$ be its line graph, whose vertex set is E . Proper edge colourings of G are proper vertex colourings of $L(G)$. An edge-colour pinning is a map $\tau: \Lambda_E \rightarrow [q]$ for some $\Lambda_E \subseteq E$; it is proper when adjacent pinned edges have distinct colours. With field variables $\lambda = (\lambda_{e,c})_{e \in E, c \in [q]}$, define

$$\widehat{Z}_G^{\text{edge}, \tau}(\lambda) := \widehat{Z}_{L(G)}^\tau(\lambda), \quad Z_G^{\text{edge}, \tau}(\lambda) := Z_{L(G)}^\tau(\lambda). \quad (41)$$

We write $Z_G^{\text{edge}} := Z_G^{\text{edge}, \emptyset}$. If τ is proper, then

$$Z_G^{\text{edge}, \tau}(\lambda) = \left(\prod_{e \in \Lambda_E} \lambda_{e, \tau_e} \right) \widehat{Z}_G^{\text{edge}, \tau}(\lambda); \quad (42)$$

otherwise $Z_G^{\text{edge}, \tau}(\lambda) \equiv 0$.

Corollary 5.4 (Edge-colour Lee–Yang polydisc). *Let q, Δ be integers with $\Delta \geq 2$ and $q \geq 3\Delta$. There exists $\theta = \theta(q, \Delta) \in (0, 1/2]$ such that, for every finite simple graph $G = (V, E)$ of maximum degree at most Δ and every edge-colour pinning τ ,*

$$\widehat{Z}_G^{\text{edge}, \tau}(\lambda) \neq 0 \quad \text{whenever} \quad |\lambda_{e,c} - 1| < \theta \quad (e \in E \setminus \Lambda_E, c \in [q]).$$

In particular, $Z_G^{\text{edge}}(\lambda) \neq 0$ in the same polydisc. For a proper pinning, $Z_G^{\text{edge}, \tau}(\lambda) \neq 0$ when the same condition is imposed at every edge-colour pair.

Proof. Put $\Delta' := 2\Delta - 2$ and let $\mathcal{H}_\Delta$ be the class of line graphs $L(G)$ with $\Delta(G) \leq \Delta$. Every graph in $\mathcal{H}_\Delta$ has maximum degree at most Δ' , the class is closed under taking induced subgraphs, and $q \geq 3\Delta \geq \Delta' + 1$. By Proposition 5.1, applied with degree bound Δ' , it suffices to show that $\mathcal{H}_\Delta$ satisfies coupling independence at $x = 0$ with constant $C_0 = 2\Delta$.

Fix an edge-colour pinning τ of G , a free edge r , and colours a, b . As in the proof of Theorem 5.2, the two laws in Definition 2.2 arise from the list-edge-colouring instance on the free edges of G in which r has list $[q]$ and every other free edge e has list $[q]$ minus the colours of its pinned neighbours, by pinning r to a and to b . Let $\deg(e)$ be the number of free edges adjacent to e . At most $2\Delta - 2$ edges of G are adjacent to e , and each pinned one removes at most one colour from the list of e , so every list, including that of r , has size at least $\deg(e) + q - (2\Delta - 2) \geq \deg(e) + \Delta + 2$. Hence the instance is $((1 + \gamma)\Delta + 1)$ -extra in the sense of [CWZZ25], with $\gamma = 1/\Delta$, and [CWZZ25, Theorem 15] bounds $W_{1, \text{Ham}}$ between the two pinned laws, on all free edges, by $1 + 2/\gamma = 1 + 2\Delta$. Every coupling of the two laws disagrees at r when $a \neq b$, so the distance on the free edges other than r is at most 2Δ . The conclusion for $Z_G^{\text{edge}, \tau}$ follows from (42). $\square$

5.2 Log-concave Boolean Holant problems

We next apply the method to log-concave Boolean Holant problems, using only the notation needed here. A symmetric Boolean signature of arity d is specified by

$$f: \{0, \dots, d\} \longrightarrow [0, \infty),$$

where $f(k)$ is the value assigned when exactly k incident edges are selected. Such a signature is *log-concave* if its positive support is an interval and

$$f(k)^2 \geq f(k-1)f(k+1) \quad (1 \leq k < d).$$

For a graph $G = (V, E)$, signatures $\mathbf{f} = (f_v)_{v \in V}$ with $\text{arity}(f_v) = \deg_G(v)$, and edge activities $\mathbf{z} = (z_e)_{e \in E}$, put

$$Z_{G, \mathbf{f}}(\mathbf{z}) := \sum_{S \subseteq E} \prod_{v \in V} f_v(\deg_S(v)) \prod_{e \in S} z_e, \quad (43)$$

where $\deg_S(v)$ counts the selected edges incident to v . Zero locations and approximation algorithms for Holant polynomials were studied in [GLLZ19, CFF⁺22]. For log-concave signatures satisfying $f_v(0) > 0$, Chen and Gu established spectral independence and rapid sampling through recursive couplings [CG24], while He, Li, Qiu, and Zhang obtained an FPTAS [HLQZ25]. Chen, Chen, and Zhang proved a spectral-independence bound for these measures that does not depend on the maximum degree [CCZ26]; our argument uses the degree-dependent coupling input of [CG24], and Remark 5.8 shows that complex zero-free widths must depend on Δ .

5.2.1 Residual signatures and coupling independence

Lemma 5.5 (Residual Holant input). *Fix an integer $\Delta \geq 2$, a finite family $\mathcal{F}$ of log-concave symmetric Boolean signatures satisfying $f(0) > 0$ for every $f \in \mathcal{F}$, and $R > 0$. Let $\mathcal{F}_{\text{res}}$ consist of all normalized residual signatures obtained as follows: for every $f \in \mathcal{F}$ and integers $j, r \geq 0$ with $j + r \leq \text{arity}(f)$ and $f(j) > 0$, take*

$$g(k) := \frac{f(j+k)}{f(j)} \quad (0 \leq k \leq r)$$

in $\mathcal{F}_{\text{res}}$. Here j is the number of incident edges fixed to 1, and r is the number left free; the remaining incident edges are fixed to 0. This is a finite family of log-concave signatures, each with initial-interval support and $g(0) = 1$.

There is $C = C(\Delta, \mathcal{F}, R) < \infty$ with the following property. Let $H = (W, F)$ be a finite simple graph of maximum degree at most Δ , carrying at each vertex w a signature $g_w \in \mathcal{F}_{\text{res}}$ of arity $\deg_H(w)$, and let $\mathbf{x} \in [0, R]^F$. For an edge $e = uv \in F$, its normalized 0-child is obtained by deleting e and restricting the signature of each endpoint $w \in \{u, v\}$ to its first $\deg_{H-e}(w) + 1$ entries. If $g_u(1)g_v(1) > 0$, its normalized 1-child instead has endpoint signatures

$$k \mapsto \frac{g_w(k+1)}{g_w(1)} \quad (0 \leq k \leq \deg_{H-e}(w)), \quad w \in \{u, v\}.$$

All other signatures are unchanged, and every child signature belongs to $\mathcal{F}_{\text{res}}$. The normalization removes the fixed-edge factor $z_e g_u(1)g_v(1)$, so the 1-child remains defined when $x_e = 0$. If $g_u(1)g_v(1) > 0$, let $\mu_{\mathbf{x}}^{e=0}$ and $\mu_{\mathbf{x}}^{e=1}$ be the Gibbs laws of the two normalized children on $\{0, 1\}^{F \setminus \{e\}}$. Then

$$W_{1, \text{Ham}}(\mu_{\mathbf{x}}^{e=0}, \mu_{\mathbf{x}}^{e=1}) \leq C.$$

If $Z^{e=0}$ and $Z^{e=1}$ denote the corresponding normalized child partition functions, then

$$Z^{e=1}(\mathbf{x}) \leq Z^{e=0}(\mathbf{x}). \quad (44)$$

When $g_u(1)g_v(1) = 0$, the unnormalized 1-child is identically zero.

Proof. The claims about $\mathcal{F}_{\text{res}}$ follow because restriction and multiplication by a positive scalar preserve log-concavity; the assumption $f(0) > 0$ makes every resulting support an initial interval. If $g(k) = f(j+k)/f(j)$ has parameters (j, r) , then its restriction to $\{0, \dots, r'\}$ with $r' \leq r$ has parameters (j, r') , and for $g(t) > 0$ the shift $k \mapsto g(t+k)/g(t) = f(j+t+k)/f(j+t)$ has parameters $(j+t, r-t)$. Hence $\mathcal{F}_{\text{res}}$ is closed under restriction and normalized shifts, so every child signature belongs to $\mathcal{F}_{\text{res}}$. Let

$$A := \max(\{0\} \cup \{g(1) : g \in \mathcal{F}_{\text{res}}, \text{arity}(g) \geq 1\}).$$

When all free activities are positive, apply the two endpoint shifts successively, shifting each original sequence before restricting it to the remaining degree. An isolated endpoint changes only a removed scalar and requires no coupling. By [CG24, Proposition 17 and Lemma 21], applying the two endpoint comparisons as in the proof of Proposition 14 there gives the Wasserstein bound with

$$C := 2((1 + A^2 R)^\Delta - 1).$$

Here the intermediate shifted signatures remain in $\mathcal{F}_{\text{res}}$, so each of positive arity has $g(1) \leq A$, and every positive free activity is at most R . Thus each endpoint comparison contributes at most $(1 + A^2 R)^\Delta - 1$, and the triangle inequality gives the displayed value of C . This bound does not depend on a positive lower bound for the free activities. If some activities vanish, replace only those coordinates by positive values tending to zero. Each normalized child has the all-zero configuration with weight one, so both Gibbs laws converge on the finite common state space. Compactness of the coupling polytope gives a limiting coupling with the same bound.

To prove (44), note that the successive ratios of a log-concave signature g are nonincreasing on its positive support. Since $g(0) = 1$, multiplying these ratios gives

$$\frac{g(t+k)}{g(t)} \leq g(k) \quad (g(t) > 0, \quad t+k \leq \text{arity}(g)). \quad (45)$$

If the numerator vanishes, the inequality is immediate. Taking $t = 1$, each normalized 1-child endpoint signature is pointwise at most its 0-child counterpart. The termwise comparison over subsets of $F \setminus \{e\}$ proves (44). $\square$

5.2.2 Zero-freeness near the nonnegative orthant

Theorem 5.6 (Log-concave Holant polytube). *Fix an integer $\Delta \geq 2$ and a finite family $\mathcal{F}$ of log-concave symmetric Boolean signatures satisfying $f(0) > 0$ for every $f \in \mathcal{F}$. For every $R > 0$, there is $\varepsilon = \varepsilon(\Delta, \mathcal{F}, R) > 0$ such that the following holds. For every finite simple graph $G = (V, E)$ of maximum degree at most Δ , every assignment $(f_v)_{v \in V}$ with $f_v \in \mathcal{F}$ of arity $\deg_G(v)$, and every $\mathbf{z} \in \mathbb{C}^E$,*

$$Z_{G, \mathbf{f}}(\mathbf{z}) \neq 0 \quad \text{whenever} \quad \mathbf{z} \in \mathcal{U}_\varepsilon([0, R])^E.$$

The activities may be perturbed independently. In particular, the union $\bigcup_{R>0} \mathcal{U}_\varepsilon(\Delta, \mathcal{F}, R)([0, R])^E$ of these polytubes, each with one R for all coordinates, is an open zero-free neighbourhood of $[0, \infty)^E$. On the diagonal $z_e = z$, the univariate specialization $Z_{G, \mathbf{f}}(z\mathbf{1})$ is nonzero throughout the open set

$$U_{\Delta, \mathcal{F}} := \bigcup_{R>0} \mathcal{U}_\varepsilon(\Delta, \mathcal{F}, R)([0, R]) \supset [0, \infty),$$

which depends only on $(\Delta, \mathcal{F})$ and is common to all the graphs and signature assignments above.

Proof. Residual instances. Let $\mathcal{F}_{\text{res}}$ be the residual family constructed in Lemma 5.5 for the present $\Delta, \mathcal{F}$. We prove the stronger assertion for every normalized instance on a graph of maximum degree at most Δ whose signatures belong to $\mathcal{F}_{\text{res}}$. Call an assignment of edge states *structurally feasible* if its signature product is nonzero, the activities being treated as indeterminates. This class of normalized instances is closed under fixing edges to structurally feasible states: removing the fixed activity factors and normalizing each shifted signature by its value at 0 again gives an instance with signatures in $\mathcal{F}_{\text{res}}$, as in Lemma 5.5. Dividing each original f_v by $f_v(0) > 0$ puts the instance in this class and only multiplies its partition function by a positive constant.

Shell decomposition and exterior response. The interaction graph of the edge variables, namely the line graph of the underlying graph, has maximum degree at most $\Delta' := 2\Delta - 2$. Let C be the constant from Lemma 5.5, take an integer $L \geq \max\{1, 64C\}$, and set

$$s := \Delta'(\Delta' - 1)^{L-1}, \quad N := 1 + \Delta' \sum_{k=0}^{L-1} (\Delta' - 1)^k.$$

Choose $0 < \alpha < \pi/4$ with $s\alpha \leq 1/16$. Up to isomorphism and apart from their activities, there are only finitely many connected normalized instances with at most N edges. Compactness over $[0, R]^N$ therefore gives a common local radius on which their partition functions are nonzero and the normalized child-response quotients in (46) below have logarithms of modulus at most α .

We now prove simultaneously, by strong induction on the number of free edges, the following two assertions for every normalized instance $H = (W, F)$, every $\mathbf{x} \in [0, R]^F$, and every $\mathbf{z} \in \mathbb{C}^F$ with $\max_{e' \in F} |z_{e'} - x_{e'}| < \varepsilon$: the partition function of H is nonzero at $\mathbf{z}$, and every edge $e \in F$ whose unnormalized 1-child is not identically zero satisfies

$$\left| \text{Log} \frac{Z^{e=1}(\mathbf{z})/Z^{e=1}(\mathbf{x})}{Z^{e=0}(\mathbf{z})/Z^{e=0}(\mathbf{x})} \right| \leq \alpha. \quad (46)$$

The logarithm is analytically continued from $\mathbf{z} = \mathbf{x}$. Both assertions are immediate when there is no free edge.

Fix an instance $H = (W, F)$ with $n := |F| \geq 1$, and fix a root edge $e \in F$ whose 1-child survives, that is, whose unnormalized 1-child is not identically zero. We first prove the response bound.

Components of the interaction graph not containing e give the same factors in the two children; the induction hypothesis makes them nonzero, so they cancel. For the response bound we may therefore replace H by the component containing e , relabel it as $H = (W, F)$, reset $n := |F|$, and write Z for its partition function. Writing Γ for its interaction graph, let

$$S_j(e) := \{e' \in F : \text{dist}_\Gamma(e, e') = j\}, \quad 1 \leq j \leq L.$$

If some $S_j(e)$ is empty, this component has at most N edges and the local compactness choice applies. Otherwise, project a coupling of the two real child laws onto these L disjoint shells. Their projected costs sum to at most C , so one shell $S = S_{j_*}(e)$ has cost at most C/L ; also $|S| \leq s$. Put

$$I := \{e' \in F \setminus \{e\} : 1 \leq \text{dist}_\Gamma(e, e') < j_*\}, \quad O := \{e' \in F : \text{dist}_\Gamma(e, e') > j_*\}.$$

For $i \in \{0, 1\}$, let $\Xi_i \subseteq \{0, 1\}^S$ be the shell assignments admitting a structurally feasible completion in the i -child. Initial-interval support makes each Ξ_i downward closed and ensures that it contains $\mathbf{0}$. The termwise comparison in Lemma 5.5 also gives $\Xi_1 \subseteq \Xi_0$; we use the common domain $\Xi := \Xi_0$.

To identify the exterior explicitly, let W_O be the vertices incident to O , and let $t_w(\xi)$ count selected shell edges at w . No vertex in W_O is incident to e or I , by the definition of the line-graph shells. For $\xi \in \Xi$, the normalized exterior signatures are therefore

$$g_{w,\xi}(k) = \frac{g_w(t_w(\xi) + k)}{g_w(t_w(\xi))} \quad (w \in W_O, \quad 0 \leq k \leq \deg_O(w)).$$

Structural feasibility and initial-interval support make every denominator positive. These signatures belong to $\mathcal{F}_{\text{res}}$ and are independent of the root state. Write Z_ξ for their partition function on O . Sum over I , absorbing the shell monomial and the removed factors $g_w(t_w(\xi))$ into $D_{i,\xi}$, to obtain

$$Z^{e=i}(\mathbf{z}) = \sum_{\xi \in \Xi_i} D_{i,\xi}(\mathbf{z}) Z_\xi(\mathbf{z}), \quad i \in \{0, 1\}.$$

Structurally infeasible assignments are absent from the sum; an assignment selecting an edge with zero base activity remains in the sum and merely has zero real weight.

By induction, $Z_\xi(\mathbf{z}) \neq 0$, so define

$$h_\xi := \text{Log} \frac{Z_\xi(\mathbf{z})}{Z_\xi(\mathbf{x})} \quad (\xi \in \Xi).$$

Write $h(\xi) := h_\xi$. If $\xi, \xi' \in \Xi$ differ in one coordinate, fix the root and every edge of I to zero, and all other shell edges to their common states. Leaving only the changed shell edge and the exterior free gives a normalized instance with $|O| + 1 < n$ free edges. Both assignments remain structurally feasible when the interior is set to zero, by initial-interval support; the two normalized children of this instance at the changed shell edge are exactly $Z_\xi, Z_{\xi'}$. Both logarithm branches vanish at $\mathbf{z} = \mathbf{x}$, so uniqueness of analytic continuation identifies $h_\xi - h_{\xi'}$ with the inductive response logarithm. Any two elements $\xi, \xi' \in \Xi$ can be joined through their coordinatewise minimum $\xi \wedge \xi'$ by a Hamming path contained in Ξ . Hence

$$|h_\xi - h_{\xi'}| \leq \alpha \text{Ham}(\xi, \xi'), \quad \max_{\xi, \xi' \in \Xi} |h_\xi - h_{\xi'}| \leq s\alpha \leq \frac{1}{16}.$$

Error control. Extend the real shell marginals by zero from Ξ_i to Ξ :

$$\nu_i(\xi) := \frac{D_{i,\xi}(\mathbf{x})Z_\xi(\mathbf{x})}{Z^{e=i}(\mathbf{x})} \quad (\xi \in \Xi_i), \quad \nu_i(\xi) := 0 \quad (\xi \notin \Xi_i).$$

Then $W_{1,\text{Ham}}(\nu_0, \nu_1) \leq C/L$. Put $M_i := \mathbb{E}_{\nu_i}[e^h]$. Adding and subtracting the local factors at $\mathbf{x}$ gives

$$\frac{Z^{e=i}(\mathbf{z})}{Z^{e=i}(\mathbf{x})} = M_i + E_i,$$

$$E_i = \frac{\sum_{\xi \in \Xi_i} (D_{i,\xi}(\mathbf{z}) - D_{i,\xi}(\mathbf{x})) Z_\xi(\mathbf{x}) e^{h_\xi}}{Z^{e=i}(\mathbf{x})}.$$

Each $D_{i,\xi}$ involves at most $|I \cup S| \leq N - 1$ variables. Only vertices touching S contribute nontrivial exterior normalization constants, so there are at most $2s$ such factors. All local data thus range over a finite family. After requiring $\varepsilon \leq 1$, there is a graph-independent $K \geq 1$ such that

$$|D_{i,\xi}(\mathbf{z}) - D_{i,\xi}(\mathbf{x})| \leq K\varepsilon \quad (i \in \{0, 1\}, \xi \in \Xi_i).$$

The all-zero interior assignment gives $D_{i,0}(\mathbf{x}) \geq 1$, whence $Z^{e=i}(\mathbf{x}) \geq Z_0(\mathbf{x})$. Repeated pointwise use of (45) on the exterior gives $Z_\xi(\mathbf{x}) \leq Z_0(\mathbf{x})$. Enlarging K to absorb $|\Xi|$ and $e^{1/16}$, we obtain

$$|E_i| \leq K\varepsilon |e^{h_0}|, \quad \left| \frac{M_i}{e^{h_0}} - 1 \right| \leq e^{1/16} - 1, \quad |M_i| \geq \frac{3}{4} |e^{h_0}|.$$

These bounds use only structural feasibility, not positivity of the base activities, and are therefore uniform on the closed box $[0, R]^F$.

Apply Lemma 3.1 on Ξ , with $A = \alpha$, $d = \text{Ham}$, and no local perturbation. It gives

$$\left| \text{Log} \frac{M_1}{M_0} \right| \leq \frac{2\alpha C}{L} \leq \frac{\alpha}{32}.$$

The preceding disc estimate fixes the same logarithm branch as continuation from $\mathbf{z} = \mathbf{x}$. For $\zeta_i := E_i/M_i$, we have $|\zeta_i| \leq 4K\varepsilon/3$. Require $\varepsilon < \min\{3/(8K), 3\alpha/(32K)\}$. Then $|\zeta_i| < 1/2$, and $|\text{Log}(1 + \zeta_i)| \leq 2|\zeta_i|$ bounds the two corrections in

$$\text{Log} \frac{M_1 + E_1}{M_0 + E_0} = \text{Log} \frac{M_1}{M_0} + \text{Log}(1 + \zeta_1) - \text{Log}(1 + \zeta_0)$$

by a total of at most $16K\varepsilon/3 < \alpha/2$. This proves (46) with room to spare.

Parent nonvanishing. Having proved the response bound for every surviving child pair, choose any edge $e = uv$ to prove nonvanishing for the current instance, and put $a_e := g_u(1)g_v(1)$. If $a_e = 0$, the unnormalized 1-child is identically zero and $Z(\mathbf{z}) = Z^{e=0}(\mathbf{z}) \neq 0$ by induction. Otherwise,

$$Z(\mathbf{z}) = Z^{e=0}(\mathbf{z}) + a_e z_e Z^{e=1}(\mathbf{z}).$$

The possible positive values of a_e form a finite set, and

$$r := \frac{Z^{e=1}(\mathbf{x})}{Z^{e=0}(\mathbf{x})} \leq 1$$

by (44). With ℓ denoting the logarithm in (46),

$$\frac{Z(\mathbf{z})}{Z^{e=0}(\mathbf{z})} = 1 + a_e x_e r e^\ell + a_e (z_e - x_e) r e^\ell.$$

The sum of the first two terms has real part at least one, whereas the last term has modulus at most $a_e e^\alpha \varepsilon$. A final uniform reduction of ε makes this modulus smaller than one and closes the induction.

For a point in the stated polytube, choose $x_e \in [0, R]$ with $|z_e - x_e| < \varepsilon$ for every edge and apply the induction. Taking the union over $R > 0$ gives the asserted open neighbourhood of $[0, \infty)^E$. On the diagonal, a single such R works for all coordinates, giving the asserted univariate domain. $\square$

5.2.3 b -matchings and b -edge covers

Corollary 5.7 (b -matchings). *Fix an integer $\Delta \geq 2$ and $R > 0$. There is $\varepsilon = \varepsilon(\Delta, R) > 0$ such that, for every finite simple graph $G = (V, E)$ of maximum degree at most Δ , every integer vector $\mathbf{b} = (b_v)_{v \in V}$ with $0 \leq b_v \leq \deg_G(v)$, and every $\mathbf{z} \in \mathbb{C}^E$,*

$$Z_{G, \mathbf{b}}^{\text{match}}(\mathbf{z}) := \sum_{\substack{S \subseteq E \\ \deg_S(v) \leq b_v \ \forall v}} \prod_{e \in S} z_e \neq 0$$

whenever $\mathbf{z} \in \mathcal{U}_\varepsilon([0, R]^E)$. Thus $\bigcup_{R>0} \mathcal{U}_{\varepsilon(\Delta, R)}([0, R]^E)$ is an open zero-free neighbourhood of $[0, \infty)^E$.

Proof. Apply Theorem 5.6 to the finite family $f_{d,b}(k) = \mathbf{1}_{[k \leq b]}$ for integers $0 \leq b \leq d \leq \Delta$ and $0 \leq k \leq d$. $\square$

Remark 5.8 (Dependence on the maximum degree). For ordinary matchings on the star $K_{1,\Delta}$, the diagonal partition function is $1 + \Delta z$. Its zero at $-1/\Delta$ shows that the width in Corollary 5.7 must satisfy $\varepsilon(\Delta, R) \leq 1/\Delta$. Thus dependence on the maximum degree is unavoidable for these neighbourhoods of $[0, R]$. For ordinary matchings much more is known: by the Heilmann–Lieb theorem [HL72], the diagonal matching partition function $\sum_M z^{|M|}$, summed over the matchings M of a finite graph G , has only real nonpositive zeros, and Wagner [Wag09] developed a general method for zero-free regions of Heilmann–Lieb type for spanning subgraphs with degree constraints. Corollary 5.7 allows arbitrary capacities b_v and gives a neighbourhood of the whole box $[0, R]^E$, including the interpolation anchor $\mathbf{0}$.

Corollary 5.9 (b -edge covers). *Fix an integer $\Delta \geq 2$ and $0 < \lambda_- \leq \lambda_+ < \infty$. There is $\varepsilon = \varepsilon(\Delta, \lambda_-) > 0$ such that, for every finite simple graph $G = (V, E)$ of maximum degree at most Δ , every integer vector $\mathbf{b} = (b_v)_{v \in V}$ with $0 \leq b_v \leq \deg_G(v)$, and every $\mathbf{z} \in \mathbb{C}^E$,*

$$Z_{G, \mathbf{b}}^{\text{cover}}(\mathbf{z}) := \sum_{\substack{S \subseteq E \\ \deg_S(v) \geq b_v \ \forall v}} \prod_{e \in S} z_e \neq 0$$

whenever $\mathbf{z} \in \mathcal{U}_\varepsilon([\lambda_-, \lambda_+]^E)$. Thus $\bigcup_{0 < \lambda_- \leq \lambda_+ < \infty} \mathcal{U}_{\varepsilon(\Delta, \lambda_-)}([\lambda_-, \lambda_+]^E)$ is an open zero-free neighbourhood of $(0, \infty)^E$.

Proof. For $b'_v := \deg_G(v) - b_v$, complementation of the selected edges gives

$$Z_{G, \mathbf{b}}^{\text{cover}}(\mathbf{z}) = \left(\prod_{e \in E} z_e \right) Z_{G, \mathbf{b}'}^{\text{match}}((z_e^{-1})_{e \in E}).$$

Thus ∞ is the natural interpolation anchor, corresponding to $\mathbf{0}$ in the reciprocal variables. Let ε_1 be the width $\varepsilon(\Delta, R)$ of Corollary 5.7 with $R = \lambda_-^{-1}$; it depends only on (Δ, λ_-) . Put $\varepsilon := \min\{\lambda_-/2, \lambda_-^2 \varepsilon_1/2\}$. If $|z_e - x_e| < \varepsilon$ with $x_e \in [\lambda_-, \lambda_+]$, then $|z_e| \geq x_e/2 > 0$ and

$$|z_e^{-1} - x_e^{-1}| = \frac{|z_e - x_e|}{|z_e| x_e} < \frac{2\varepsilon}{\lambda_-^2} \leq \varepsilon_1,$$

while $x_e^{-1} \in [\lambda_+^{-1}, \lambda_-^{-1}] \subseteq [0, \lambda_-^{-1}]$. Hence $\prod_e z_e \neq 0$ and $(z_e^{-1})_{e \in E} \in \mathcal{U}_{\varepsilon_1}([0, \lambda_-^{-1}])^E$, so Corollary 5.7 applies; the width depends neither on G nor on λ_+ . $\square$

The same diagonal specialization in Corollaries 5.7 and 5.9 gives common open univariate zero-free neighbourhoods of $[0, \infty)$ and $(0, \infty)$, respectively, depending only on Δ .

A Further zero-free regimes for the Potts model

This appendix records further Potts zero-free regimes obtained by combining the transfer results of the main text with additional coupling-independence bounds. The source of each coupling input is specified below.

The high-temperature row of Table A.1 follows from Proposition 4.12 and Lemma 3.2 of this paper, and the near-Vigoda row reduces by integrality to Theorem 1.1 and the Carlson–Vigoda row. The remaining coupling-independence inputs, which extend known hard-colouring couplings to positive activities or establish new ones, are proved in the companion article [SS26a].

The companion proofs of these inputs were produced by AI models — GPT-5.6 Sol, except the girth-five results, which were proved by GPT-6 Astra. A Lean formalization, developed with assistance from GPT-6 Astra and Claude Opus 5.5, proves the coupling-independence and zero-free statements for these regimes [Shi26]; their public entry points retain only the displayed mathematical hypotheses. It also proves, in the form the proofs use, the results cited in the three girth rows, [CLMM23, Lemmas 5.13 and 8.7 and Equation (10)] and [BBR25, Proposition 2.6(i) and Theorem 2.5]. In some places the Lean proof follows a different route from the written one. At the twenty near-Vigoda pairs $(q, \Delta) = (11j, 6j)$, $j \leq 20$, where the written proof cites [CFG⁺25, Theorem 20], it extends the Carlson–Vigoda contraction to $q \geq 11\Delta/6$ for every $\Delta \geq 6$; and it treats the cases $\Delta = q + 2$ of the BBR row (Bencs, Berrekkal, and Regts [BBR25]) directly. More generally, the formalization proves every numbered theorem, lemma, proposition, and corollary of this article and of [SS26a], some in an equivalent form, among them Theorems 1.1 and 1.2, the case $q = 11\Delta/6$ included, and the colour-field and Holant results of Sections 5.1 and 5.2, all without external input. Only citation-level claims and the algorithmic remarks of Section 1 are not formalized.

Table A.1 gives the additional zero-free regions. The coupling input of each row originates as follows.

- Near-Vigoda: by integrality [SS26a, Lemma 4.3], the union of Theorem 1.1 (with [CFG⁺25, Theorem 20] in the equality case $q = 11\Delta/6$, that is, $(q, \Delta) = (11j, 6j)$ with $j \geq 1$) and the Carlson–Vigoda row. The constant $1/84000$ is the value of ε_0 fixed in the proof of [CDM⁺19, Theorem 5.1].
- Carlson–Vigoda: the contraction of [CV25] (at $x = 0$, [CFG⁺25, Theorem 20]), extended to $x \in [0, 1]$ [SS26a, Theorem 5.1].
- Large-girth: the mechanism of [CLMM23] (at $x = 0$, as in Theorem 5.2), extended to $x \in [0, 1]$ [SS26a, Theorem 6.1].
- High-temperature: Proposition 4.12.
- BBR interval: the tree contraction of [BBR25], strengthened to tree total-influence decay and transferred to large-girth graphs by the mechanism of [CLMM23]; see [SS26a, Theorem 7.6].
- Edge-Potts: a slot-lift adaptation of the list-edge-colouring coupling of [CWZZ25] (whose $x = 0$ bound enters Corollary 5.4), with coupling-independence constant $\Delta - 1$ on all of $[0, 1]$, including $x = 0$ [SS26a, Theorem 8.1].
- Girth-five: a covariance recursion built on the all-pinning spectral gap of [SS26a, Theorem 9.7], converted by the sphere-to-coupling lemma [CLMM23, Lemma 5.13] into the coupling bound [SS26a, Theorem 9.4]. This setting parallels the spectral local-to-global framework of [CL26], from which no input is taken.

Regime	Parameters	Zero-free near
near-Vigoda	$\Delta \geq 2, q \geq (11/6 - 1/84000)\Delta$	$[0, 1]$
Carlson–Vigoda	$\Delta \geq 125, q \geq 1.809\Delta$	$[0, 1]$
large-girth	$\Delta \geq 3, q \geq \Delta + 3; \text{girth}(G^\tau) \geq g_*(q, \Delta)$	$[0, 1]$
high-temperature	$\Delta \geq 2, q > (11/6)(1 - x_*)\Delta, x_* \in (0, 1]$	$[x_*, 1]$
BBR interval	$q \geq 3, \Delta/q \geq (e - 1/2)/(e - 1); \text{girth}(G^\tau) \geq g_{\text{BBR}}(q, \Delta)$	$[x_0, 1]$
edge-Potts	$\Delta \geq 2, q \geq 3\Delta$	$[0, 1]$
girth-five	$q \geq (1 + \delta)\Delta, 0 < \delta \leq 1; \Delta \geq \Delta_5(\delta), \text{girth}(G^\tau) \geq 5$	$[0, 1]$

In every row the conclusion is $\widehat{Z}_G^\tau(z) \neq 0$ for all z in an ε -neighbourhood of the stated interval, with $\varepsilon > 0$ depending only on the listed parameters, uniformly over graphs and pinning. The unnormalized conclusions are specified in Remark A.1. In the edge-Potts row the polynomial is $\widehat{Z}_{L(G)}^\tau$, with τ an edge-colour pinning. The thresholds $g_*(q, \Delta)$, $g_{\text{BBR}}(q, \Delta)$, and $\Delta_5(\delta)$ are those of [SS26a, Theorems 6.1, 7.6, and 9.4]; BBR refers to Bencs, Berrekkal, and Regts [BBR25]. In the BBR row, $x_0 := 1 - \frac{q}{\Delta} \left(1 - \frac{k}{\Delta}\right)^2 \frac{\Delta - k}{\Delta - k/2}$ with $k := e^{\frac{\Delta - q/2}{\Delta - q}}$ (and $x_0 := 3/4$ if $(q, \Delta) = (3, 4)$). The girth-five row assumes no bounds on local short-cycle counts.

Table A.1: Further Potts zero-free regimes obtained from coupling independence.

Remark A.1 (Transfer and the hard endpoint). The full-interval Potts rows use Theorem 1.2 (on $L(G)$, with degree bound $2\Delta - 2$, for edge-Potts). The high-temperature and BBR rows use the positive-activity induction in Lemma 3.2. In the rows with a condition $\text{girth}(G^\tau) \geq g$ ($g = g_*, g_{\text{BBR}}$, or 5), these results are applied to $\mathcal{G}_{\Delta, g}$ after replacing (G, τ) by $(\tilde{G}, \tilde{\tau})$: starting from G^τ , for every edge up with $u \in V^\tau$ and $p \in \Lambda$, attach to u a new leaf pinned to τ_p . This preserves $\widehat{Z}_G^\tau$ and the degree in G of every free vertex, the new leaves have degree one, and $\text{girth}(\tilde{G}) = \text{girth}(G^\tau)$; hence $\tilde{G} \in \mathcal{G}_{\Delta, g}$. For rows with interval $[0, 1]$, the unnormalized pinned polynomial has no zeros in the stated neighbourhood except, when $m_G(\tau) \geq 1$, a zero of multiplicity $m_G(\tau)$ at 0 ($m_{L(G)}(\tau)$ for edge-Potts). For the two positive-interval rows, choose ε smaller than the left endpoint; the unnormalized polynomial is then nonzero throughout the same neighbourhood.

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